

Fort Pickens State Park Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

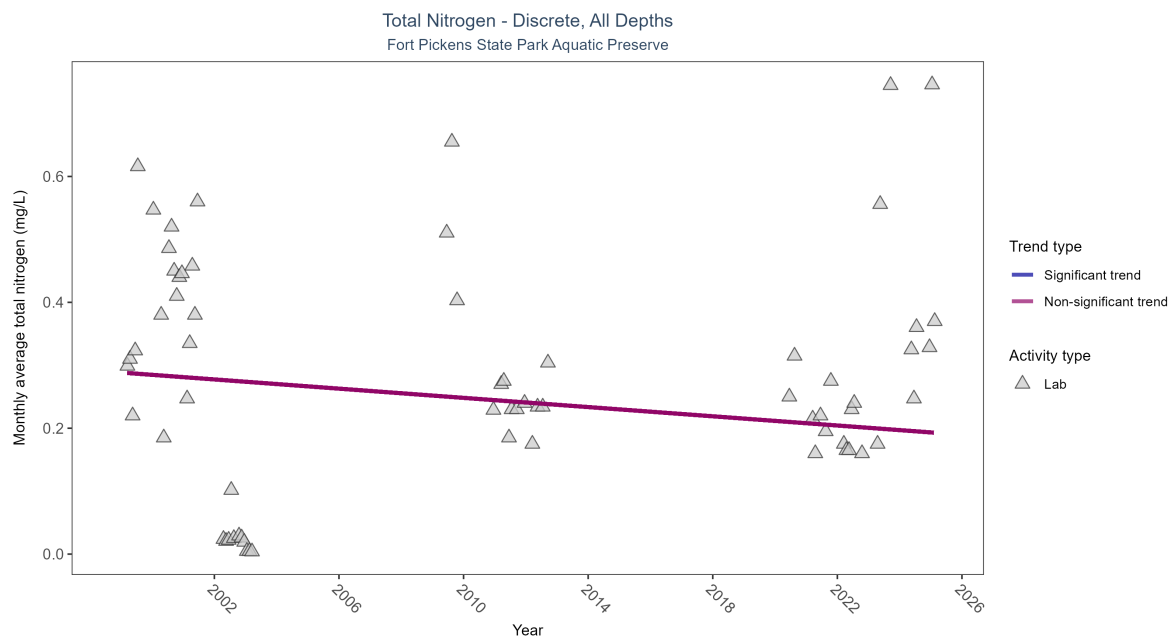


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Total nitrogen showed no detectable trend between 1999 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	119	15	1999 - 2025	0.245	-0.08647	0.28849	-0.00366	0.3742

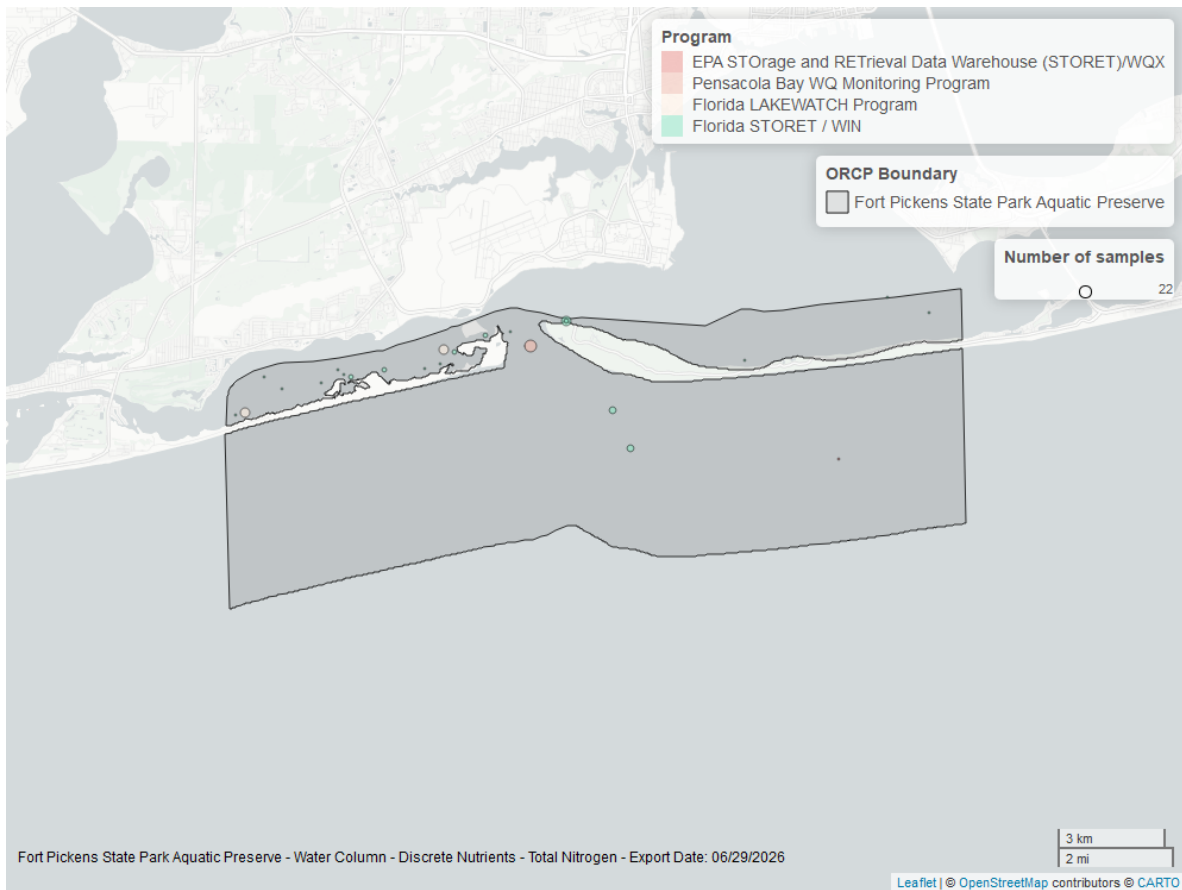


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

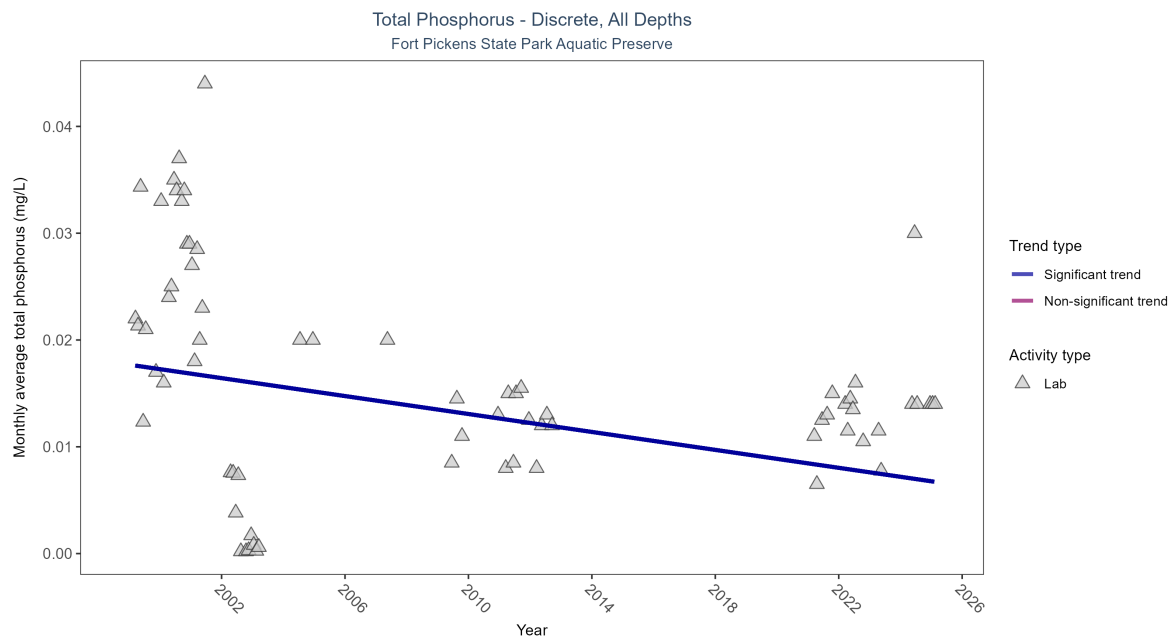


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	118	16	1999 - 2025	0.0135	-0.32037	0.01769	-0.00042	0.0072

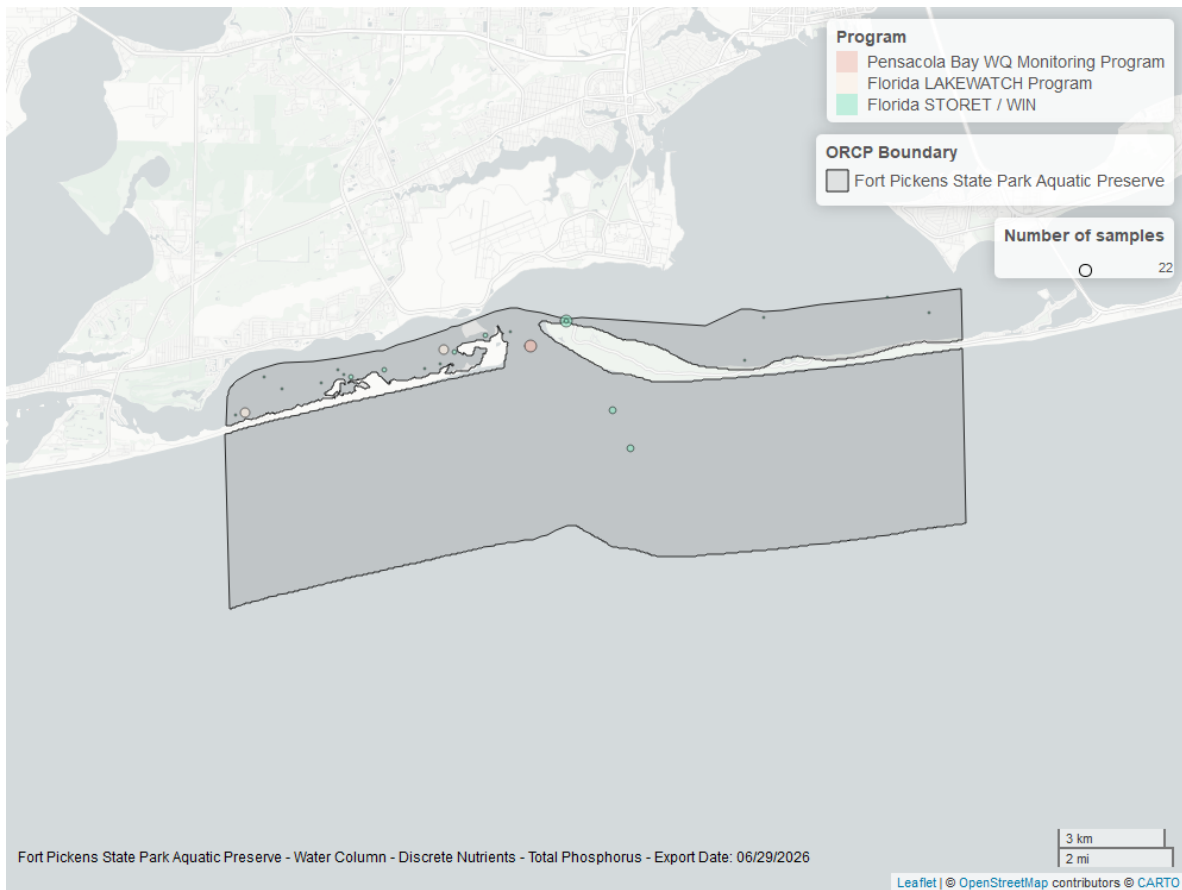


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

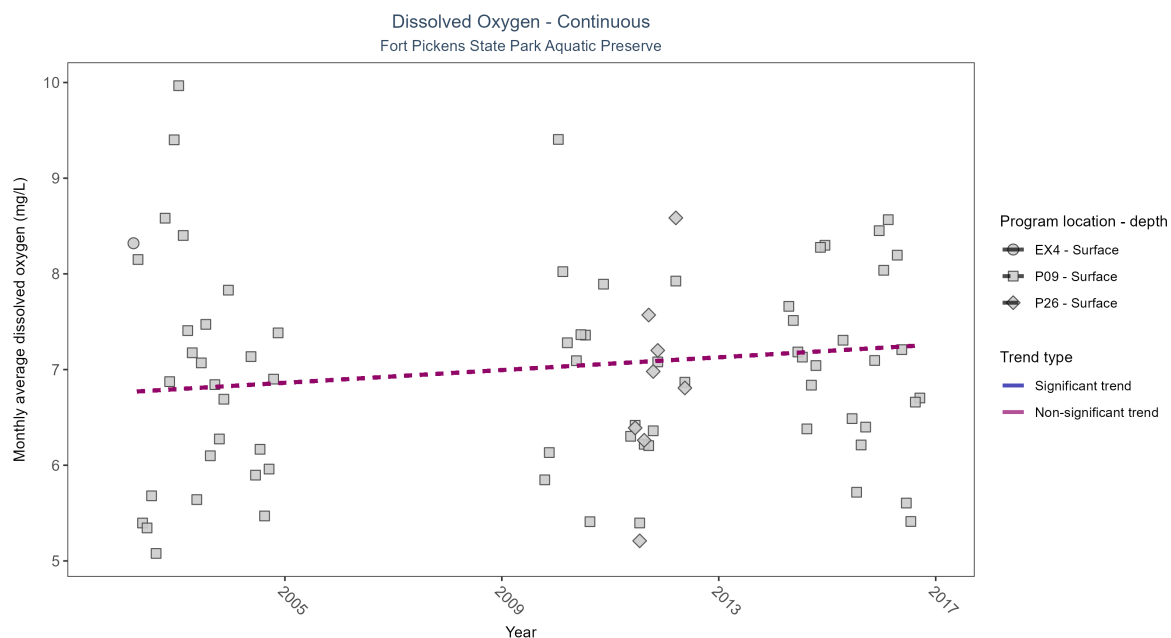


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: No detectable change in monthly average dissolved oxygen was observed at one location. There was insufficient data to fit a model for two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
EX4	Insufficient data	2	1	2002 - 2002	8.32	-	-	-	-
P09	No detectable trend	364	10	2002 - 2016	6.82	0.14	6.76	0.03	0.1
P26	Insufficient data	27	2	2011 - 2012	6.61	-	-	-	-

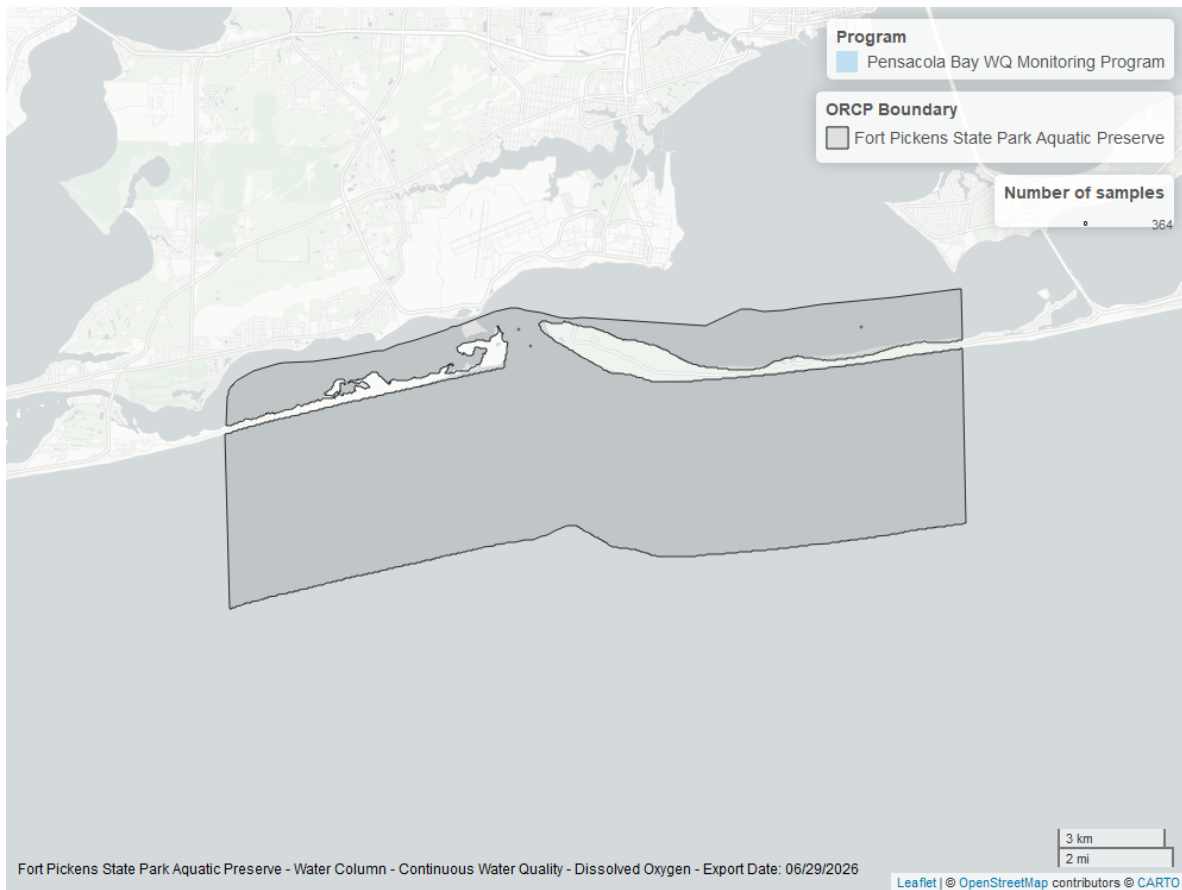


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

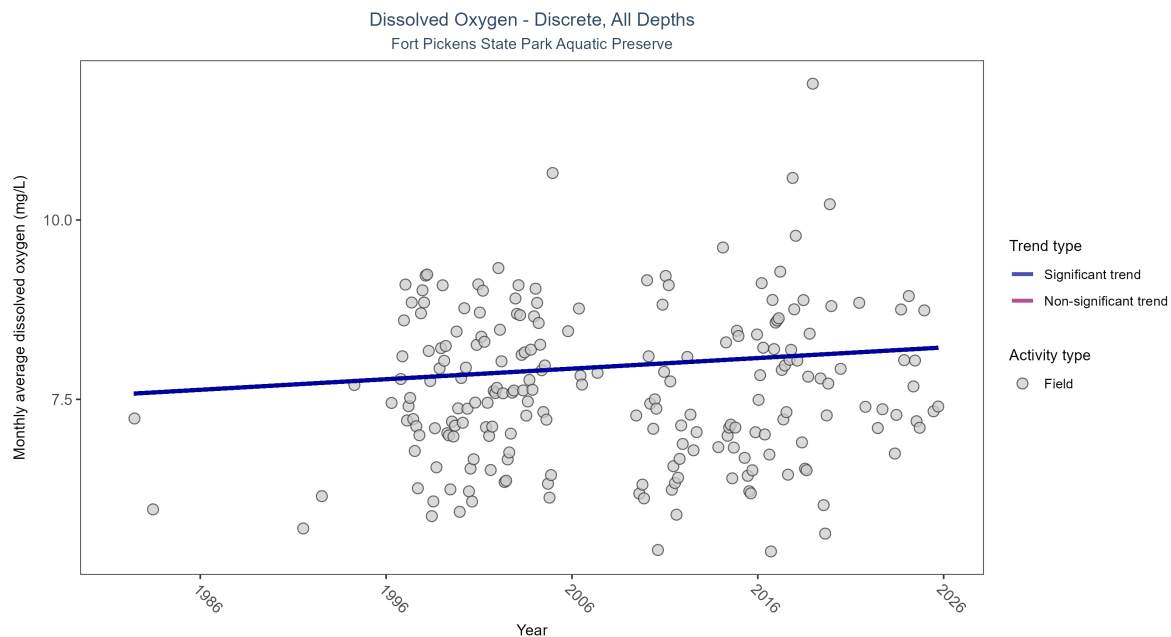


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Monthly average dissolved oxygen increased by 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	1212	34	1982 - 2025	7.505	0.1119	7.57427	0.01475	0.0193

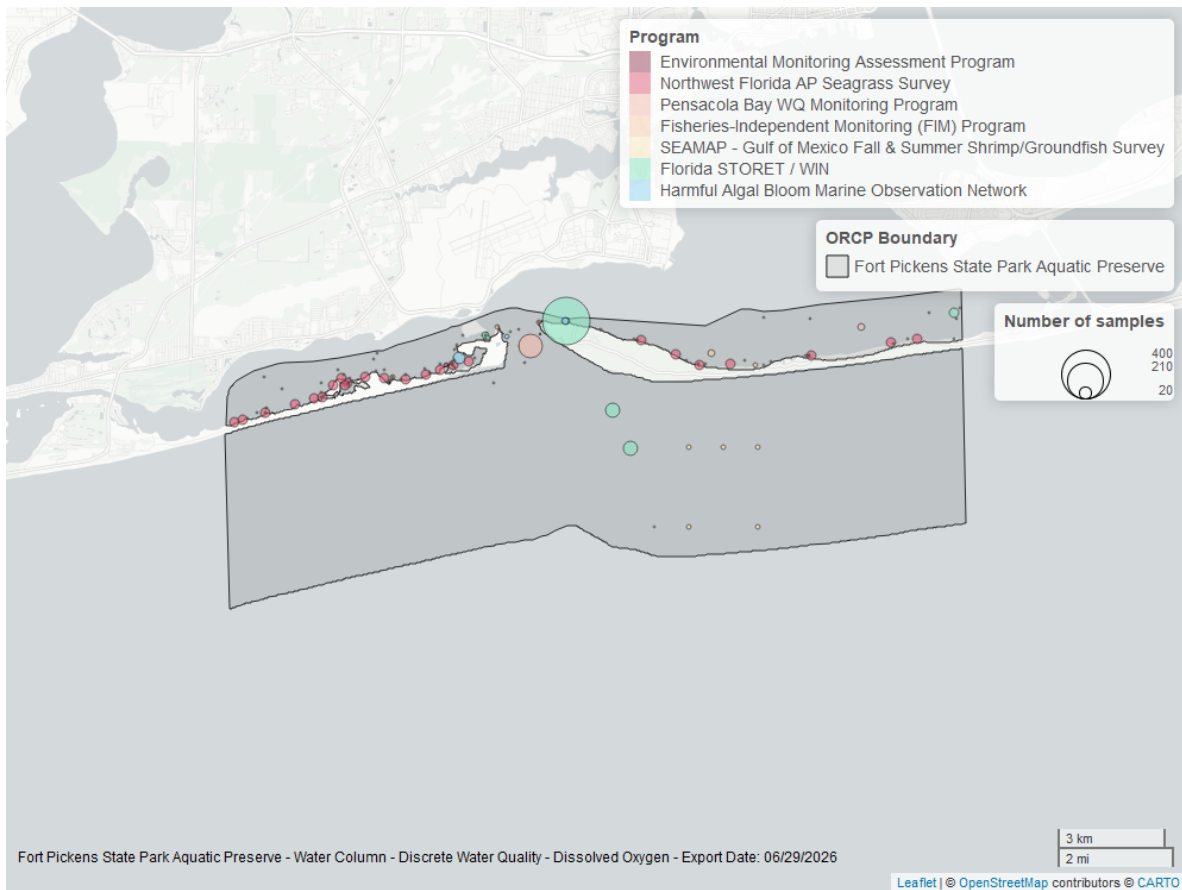


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

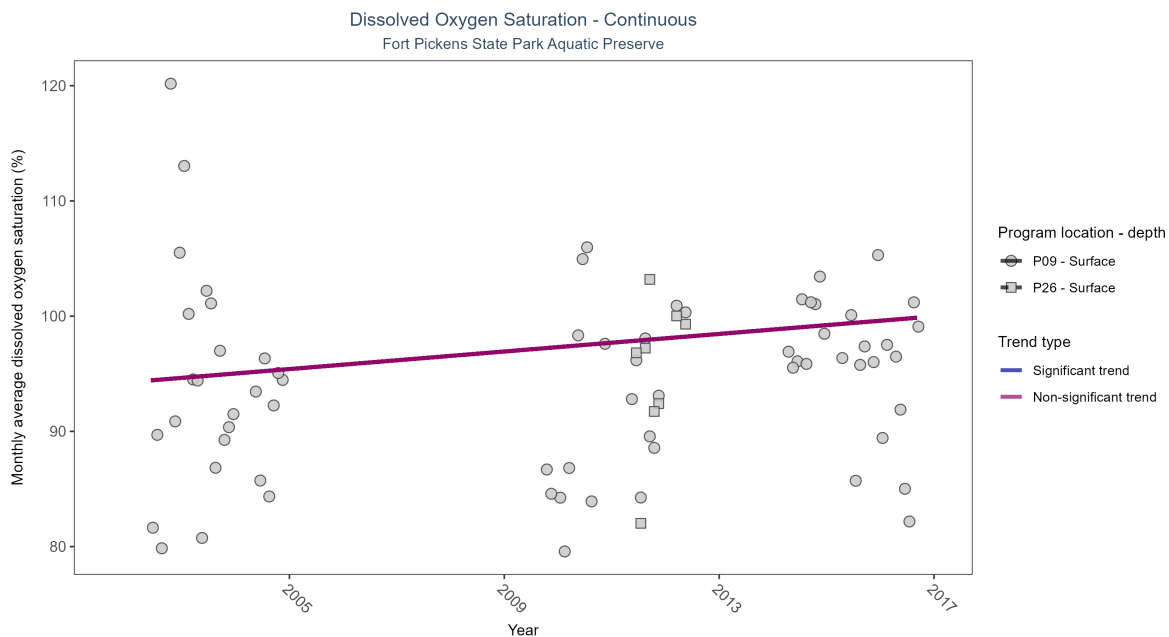


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: No detectable change in monthly average dissolved oxygen saturation was observed at one location. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
P09	No detectable trend	358	10	2002 - 2016	96.91	0.11	94.26	0.38	0.24
P26	Insufficient data	27	2	2011 - 2012	98.17	-	-	-	-

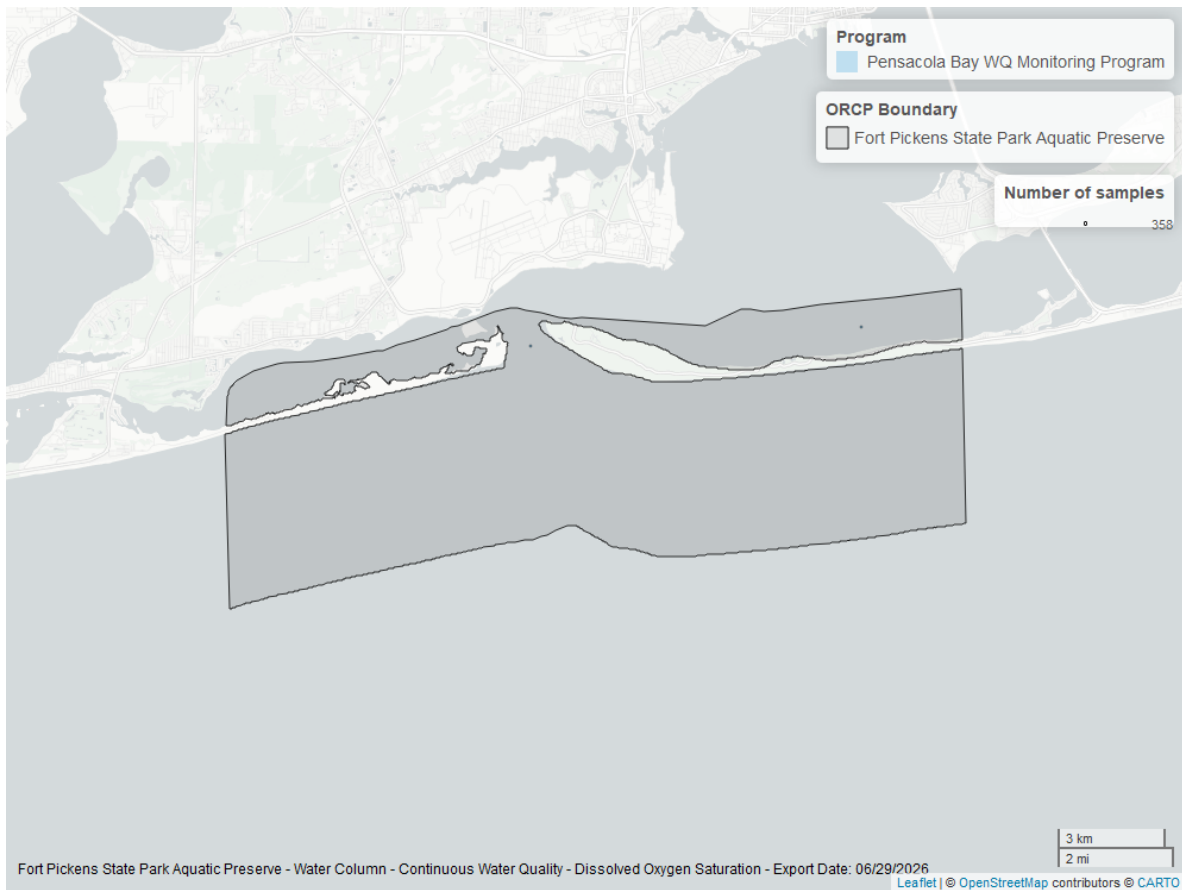


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

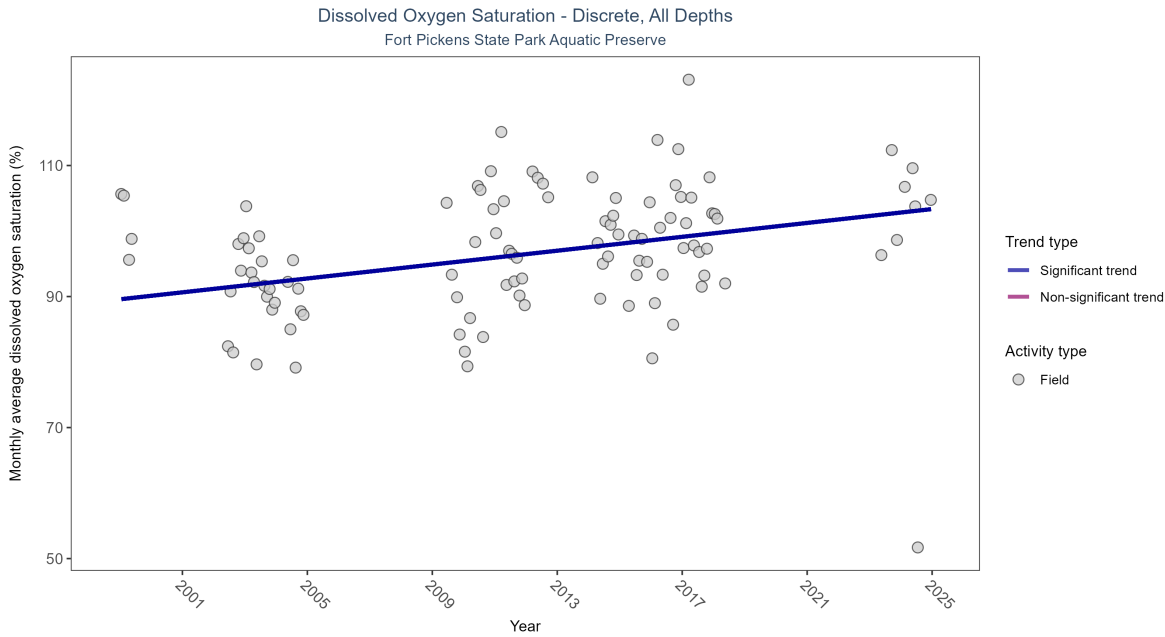


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

\begin{table}[H] \caption{Monthly average dissolved oxygen saturation increased by 0.53% per year.}

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	261	15	1999 - 2024	98.6	0.27656	89.58022	0.52972	0.0004

\end{table}

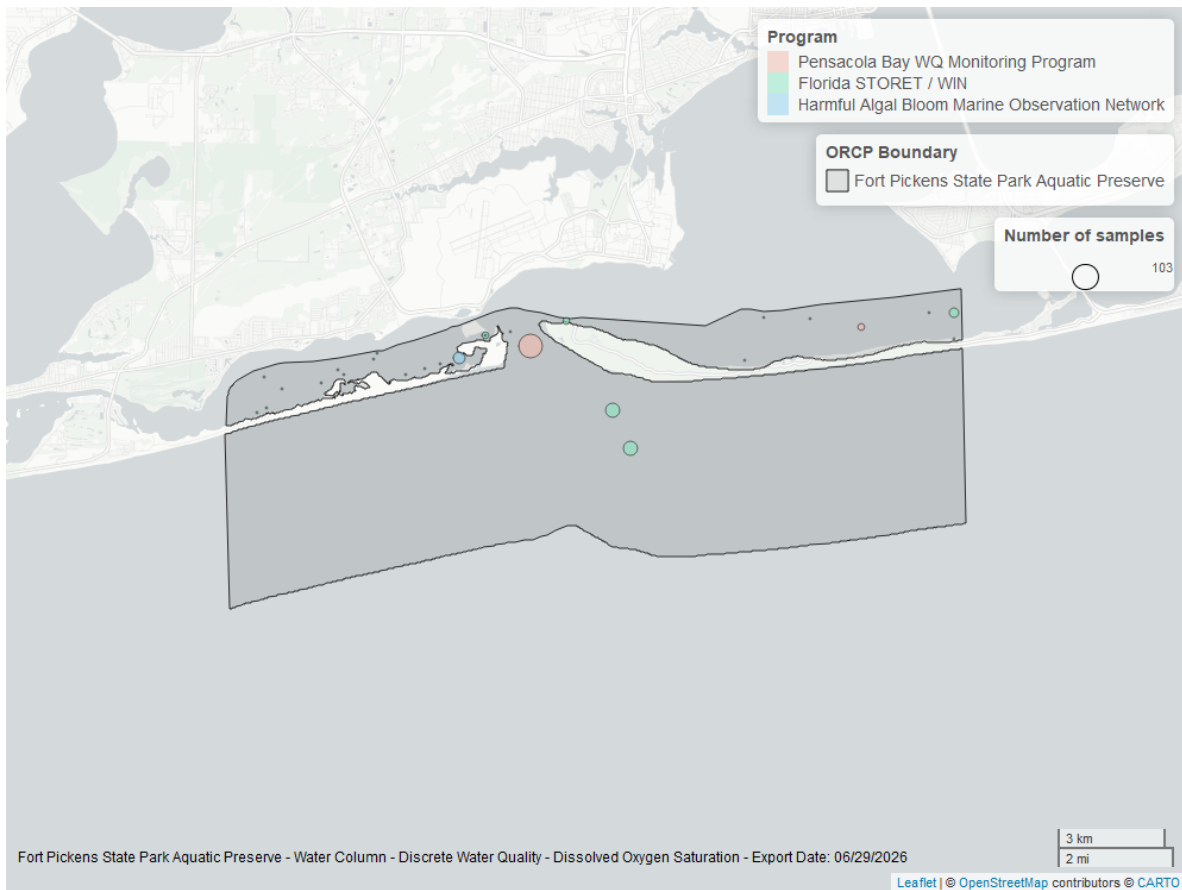


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

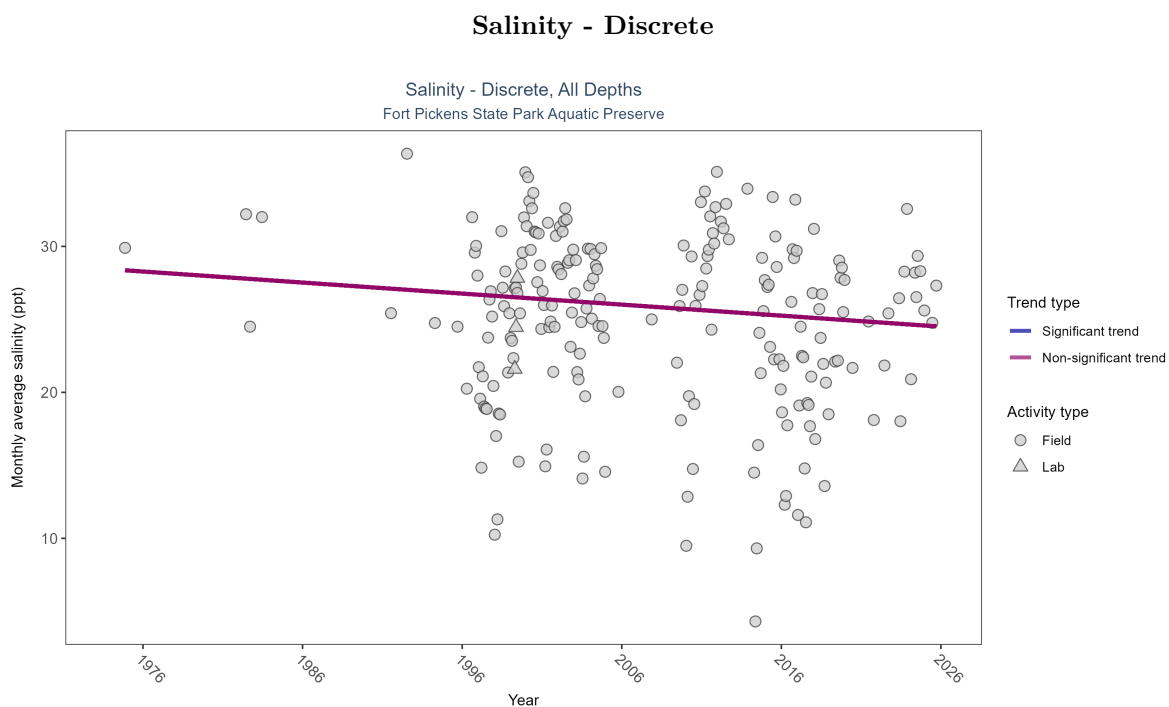


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 6: Salinity showed no detectable trend between 1974 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	No detectable trend	1230	35	1974 - 2025	26.2	-0.08405	28.42861	-0.07554	0.0824

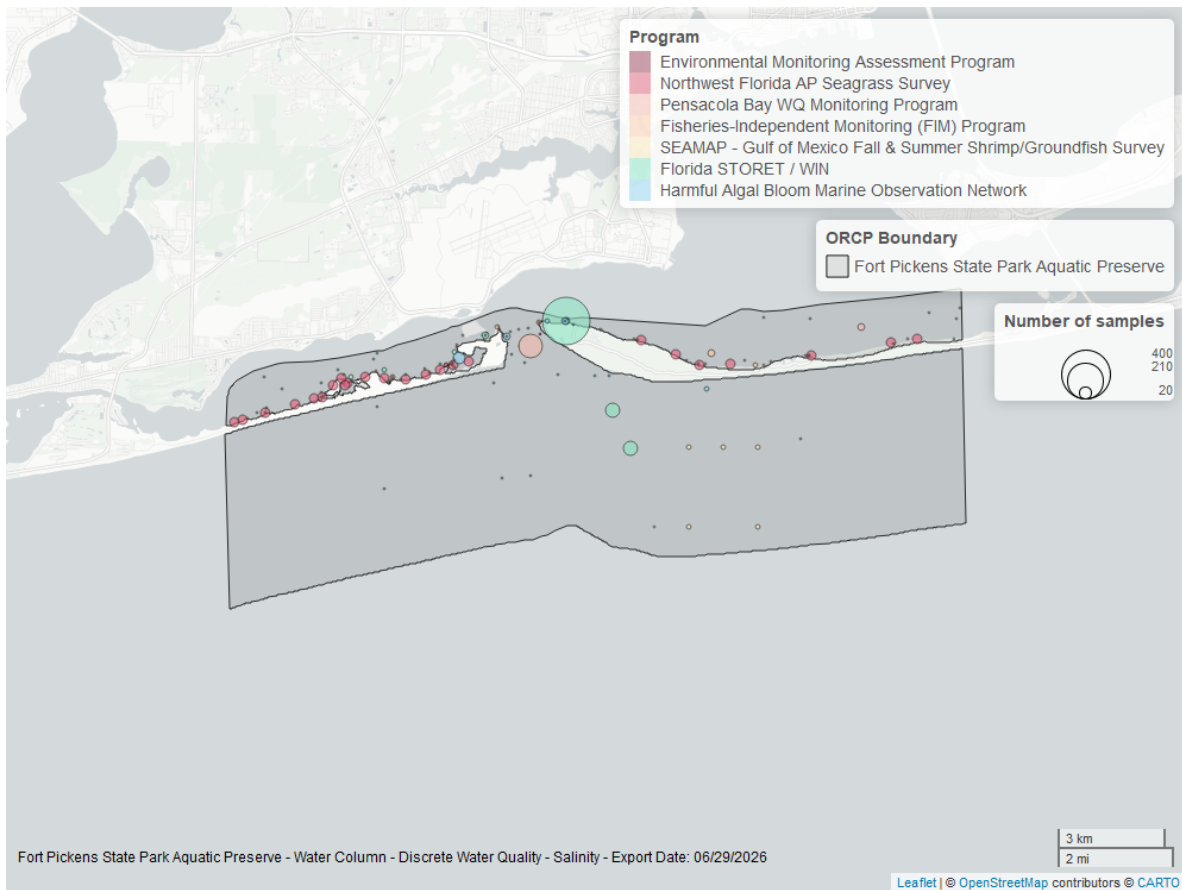


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

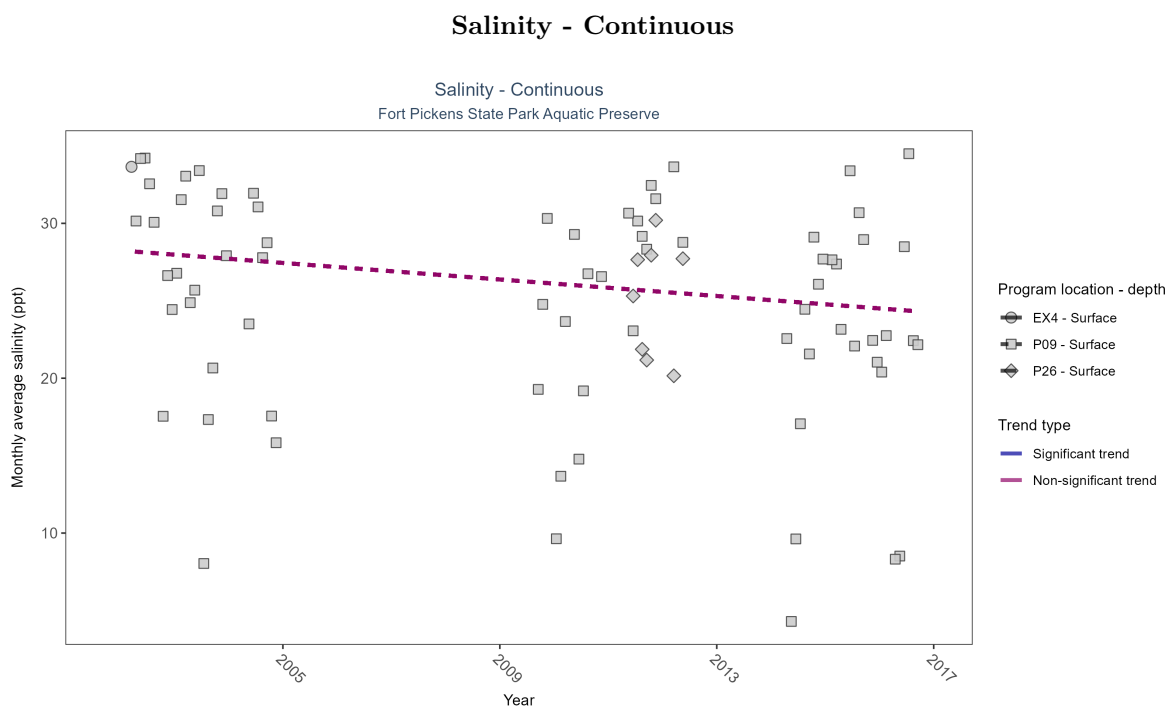


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 7: No detectable change in monthly average salinity was observed at one location. There was insufficient data to fit a model for two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
EX4	Insufficient data	2	1	2002 - 2002	33.65	-	-	-	-
P09	No detectable trend	373	10	2002 - 2016	25.91	-0.17	28.25	-0.27	0.13
P26	Insufficient data	27	2	2011 - 2012	26.38	-	-	-	-

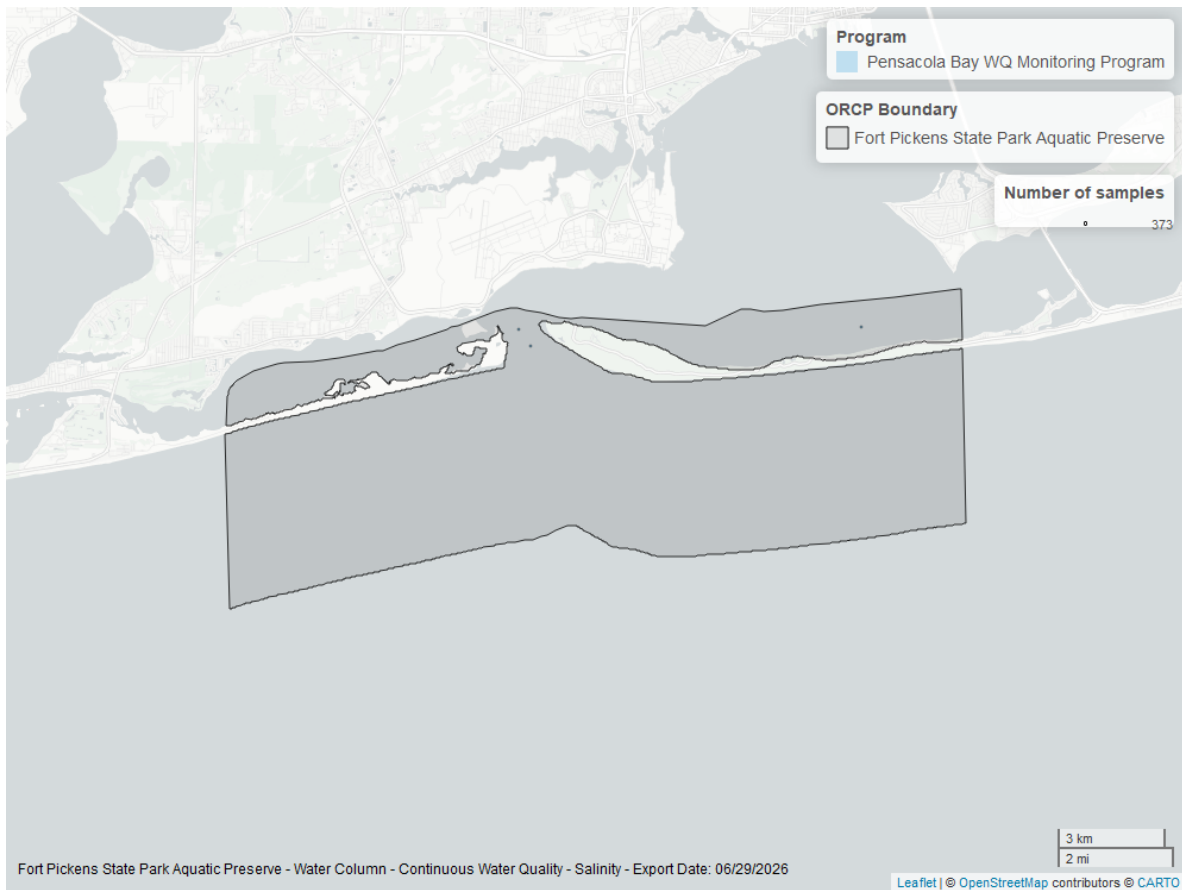


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

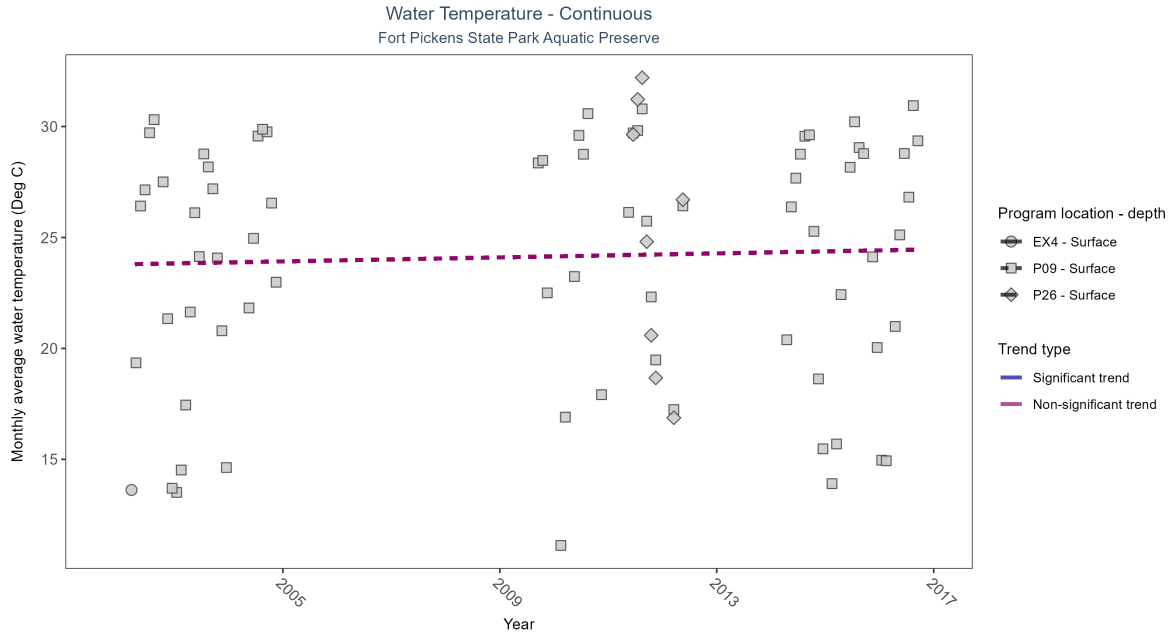


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: No detectable change in monthly average water temperature was observed at one location. There was insufficient data to fit a model for two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
EX4	Insufficient data	2	1	2002 - 2002	13.62	-	-	-	-
P09	No detectable trend	373	10	2002 - 2016	26.39	0.1	23.79	0.05	0.45
P26	Insufficient data	27	2	2011 - 2012	29.64	-	-	-	-

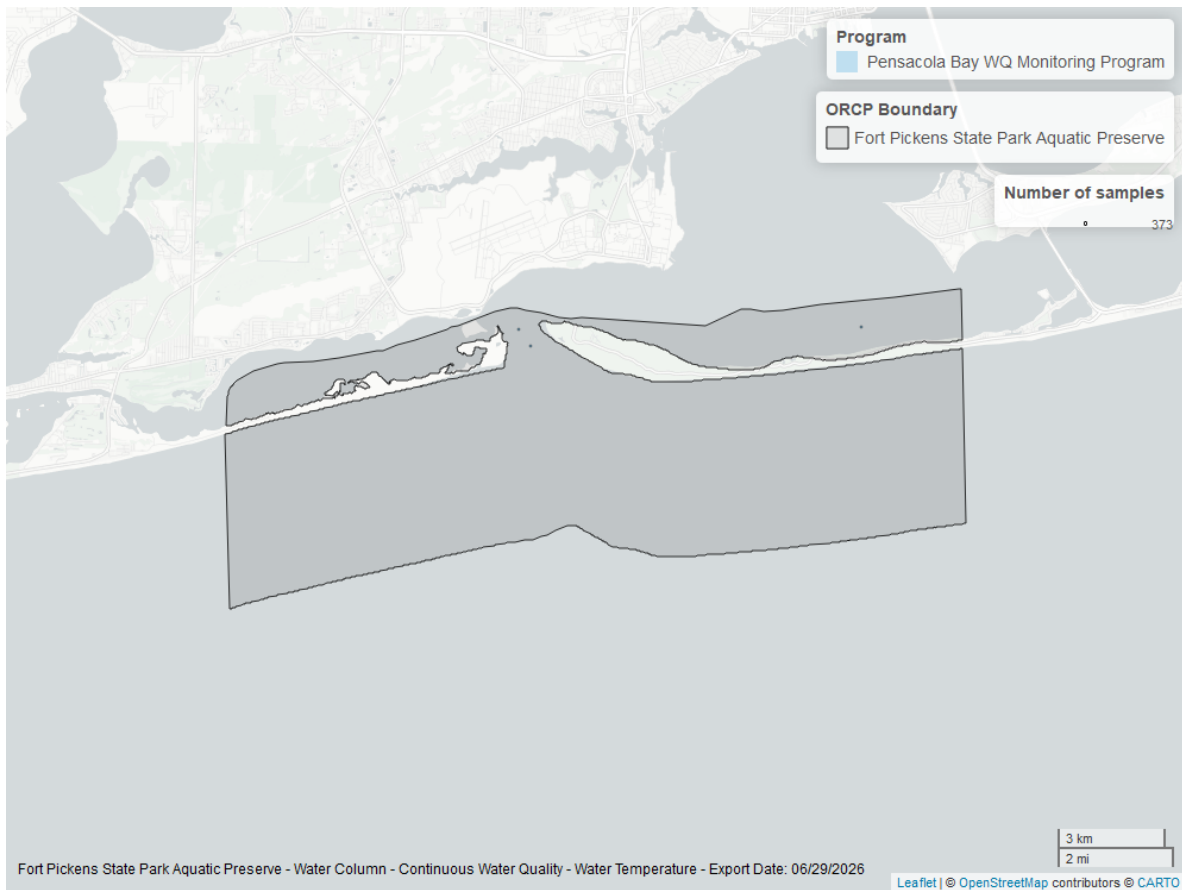


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

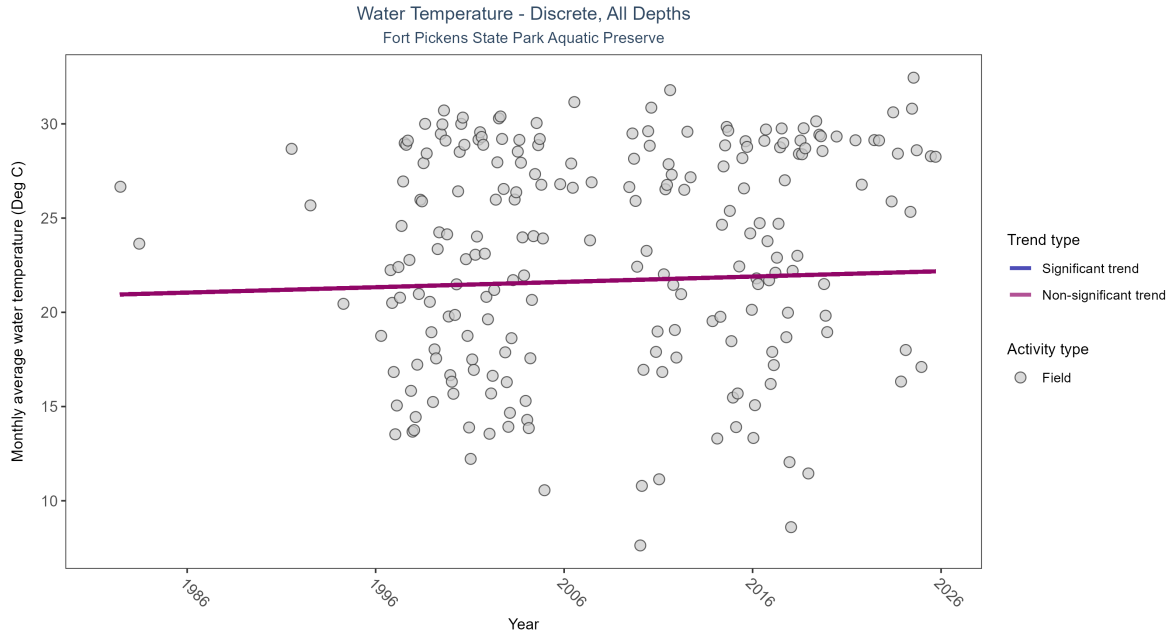


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 9: Water temperature showed no detectable trend between 1982 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	1238	34	1982 - 2025	27.4	0.08193	20.9339	0.02836	0.0517

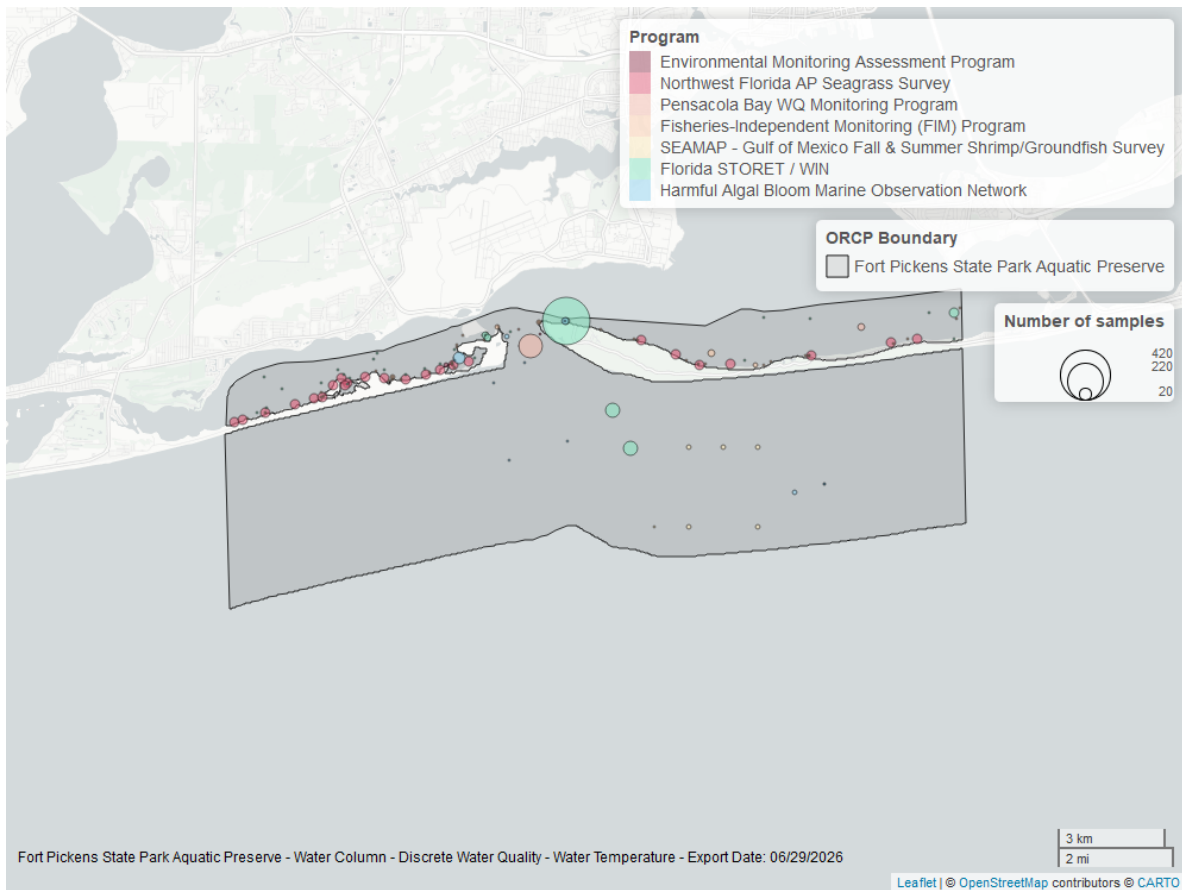


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

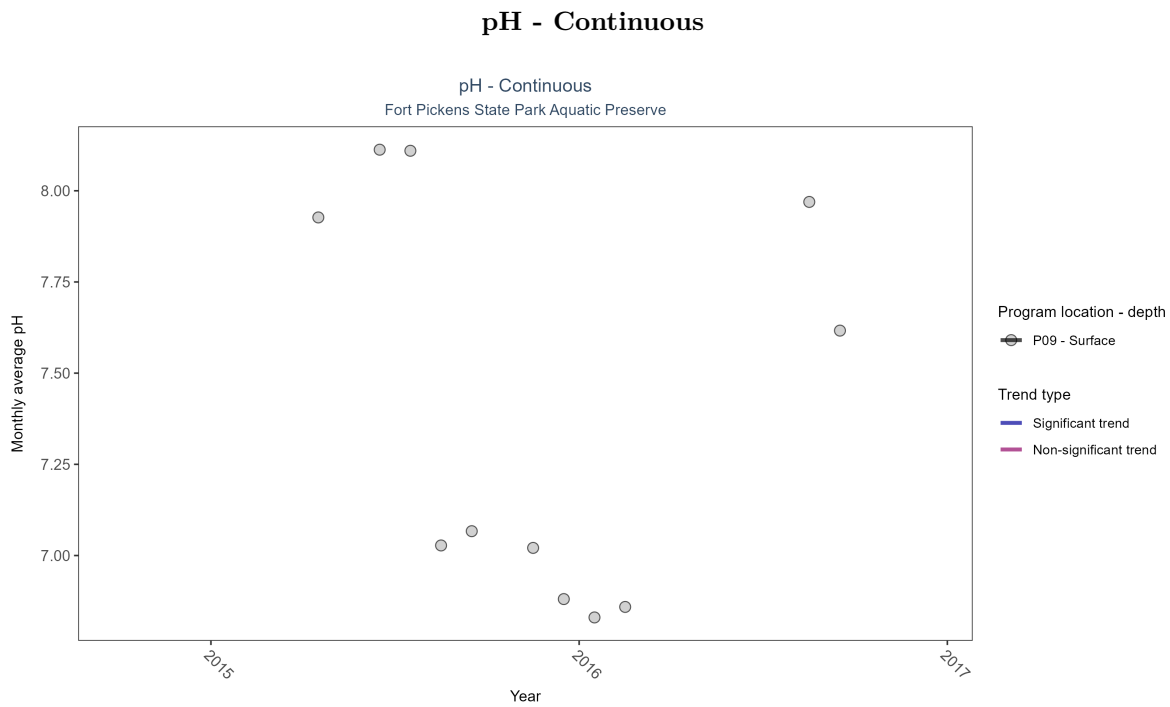


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 10: There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
P09	Insufficient data	157	2	2015 - 2016	7.06	-	-	-	-

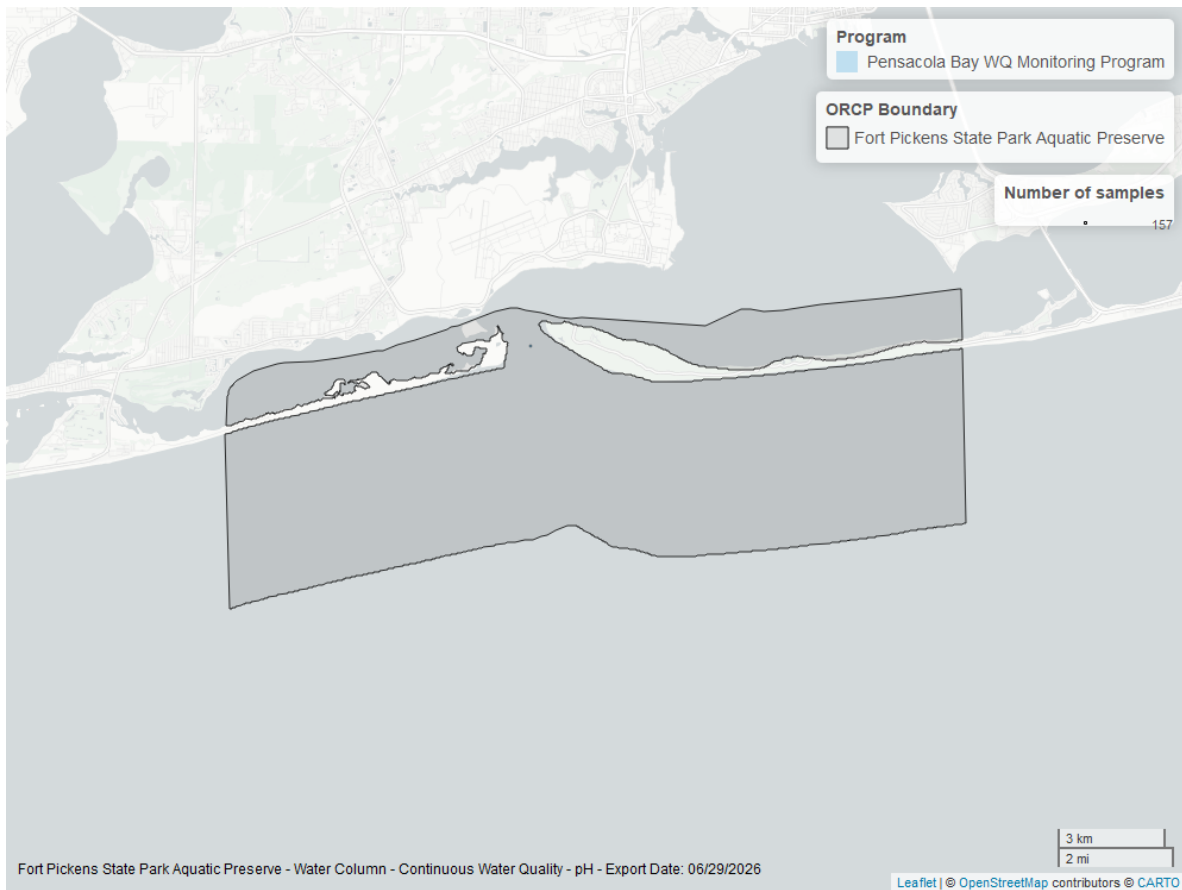


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

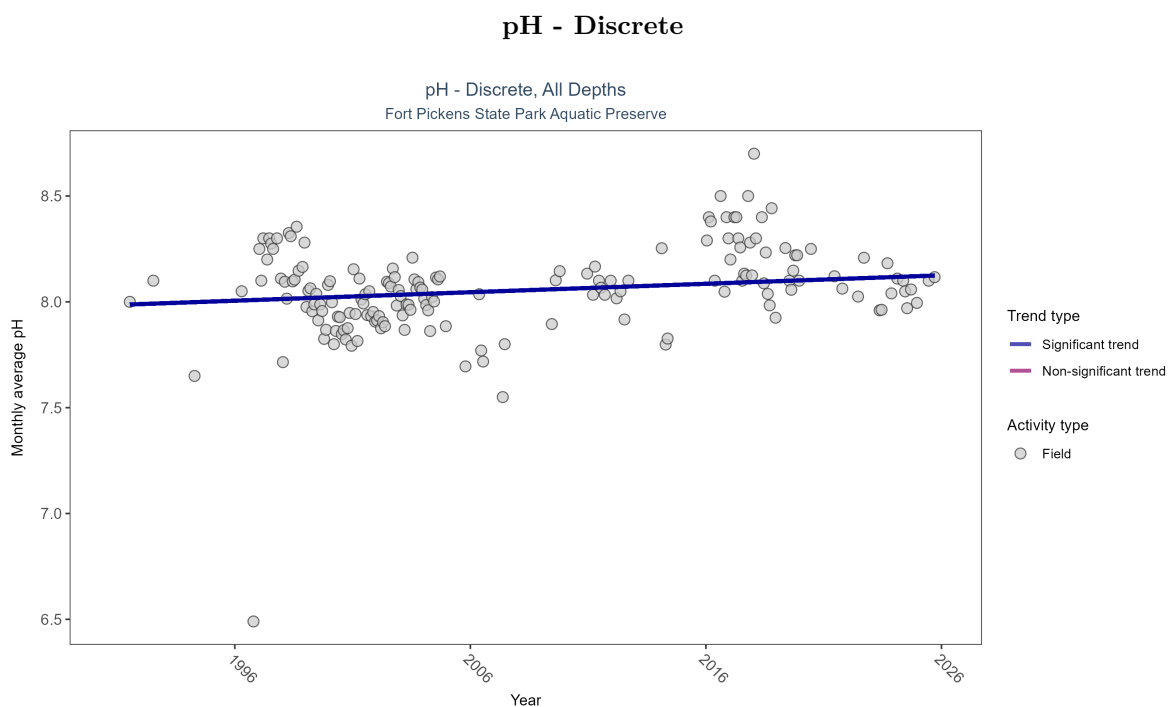


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 11: Monthly average pH increased by less than 0.01 pH units per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	1003	30	1991 - 2025	8.1	0.17298	7.98539	0.004	0.0077

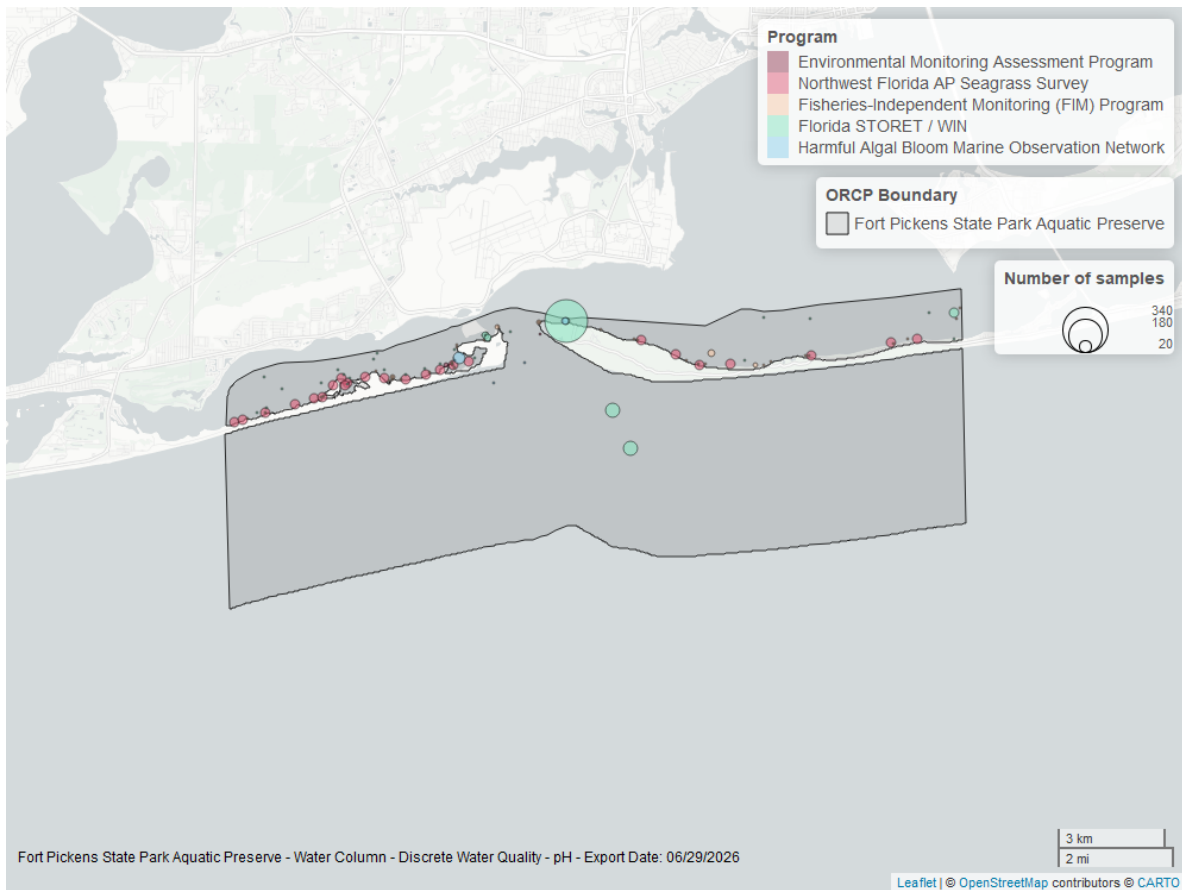


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

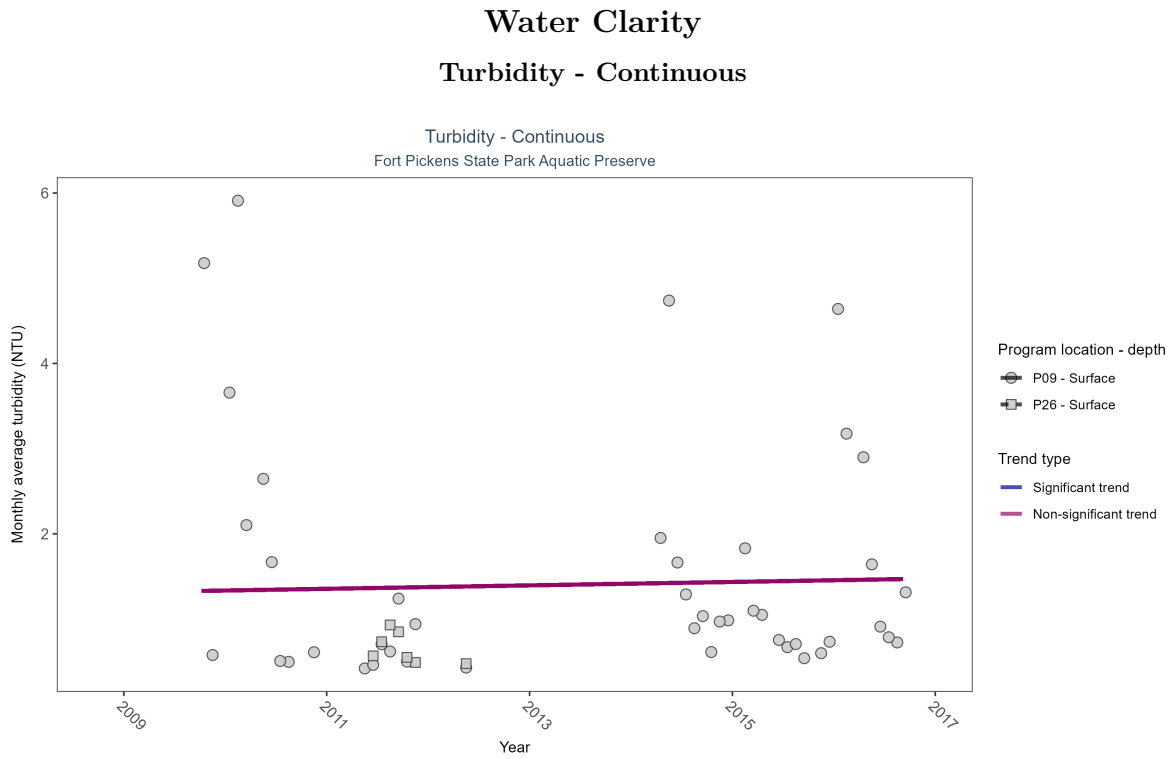


Figure 25: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 12: No detectable change in monthly average turbidity was observed at one location. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
P09	No detectable trend	275	7	2009 - 2016	0.85	0.07	1.31	0.02	0.38
P26	Insufficient data	25	2	2011 - 2012	0.62	-	-	-	-

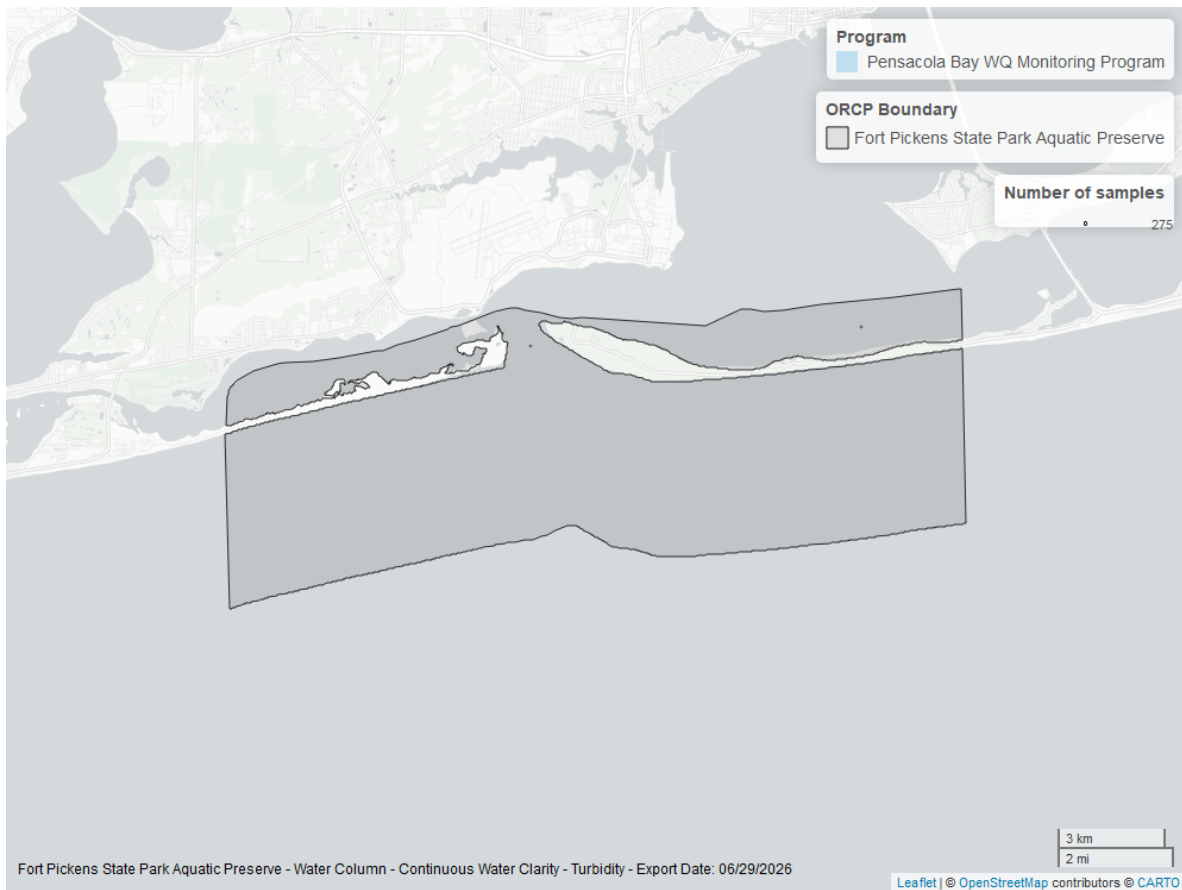


Figure 26: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

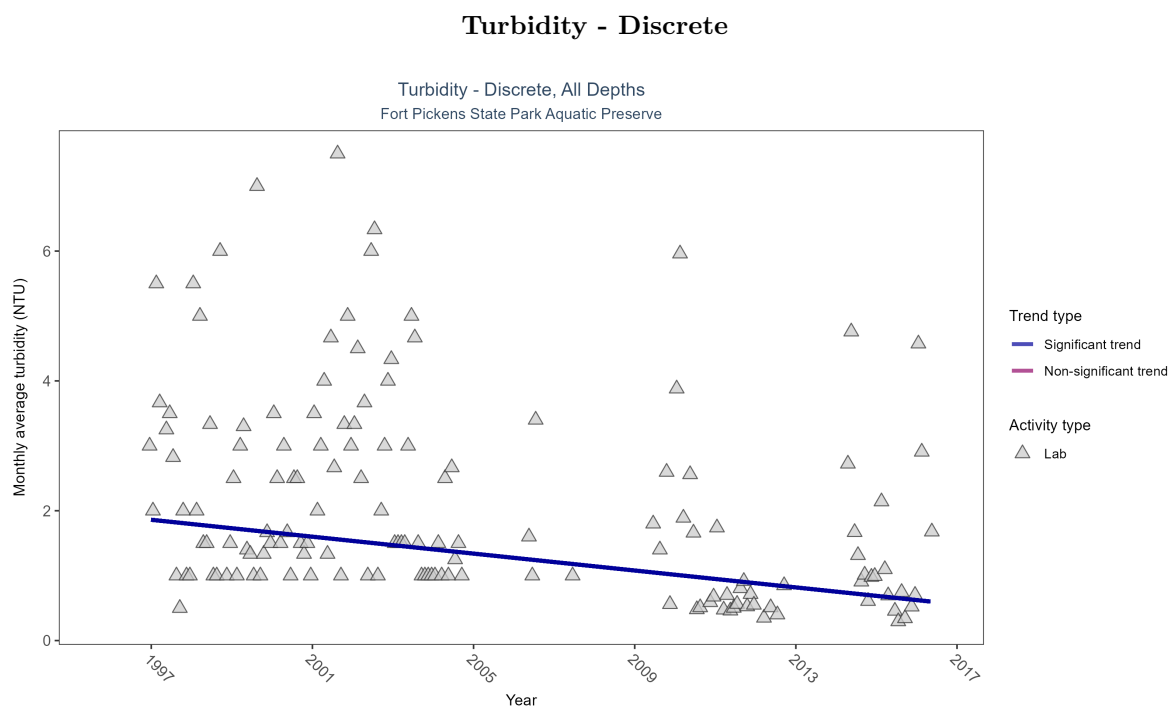


Figure 27: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 13: Monthly average turbidity decreased by 0.07 NTU per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	313	18	1996 - 2016	1.0122	-0.31425	1.92558	-0.06511	0

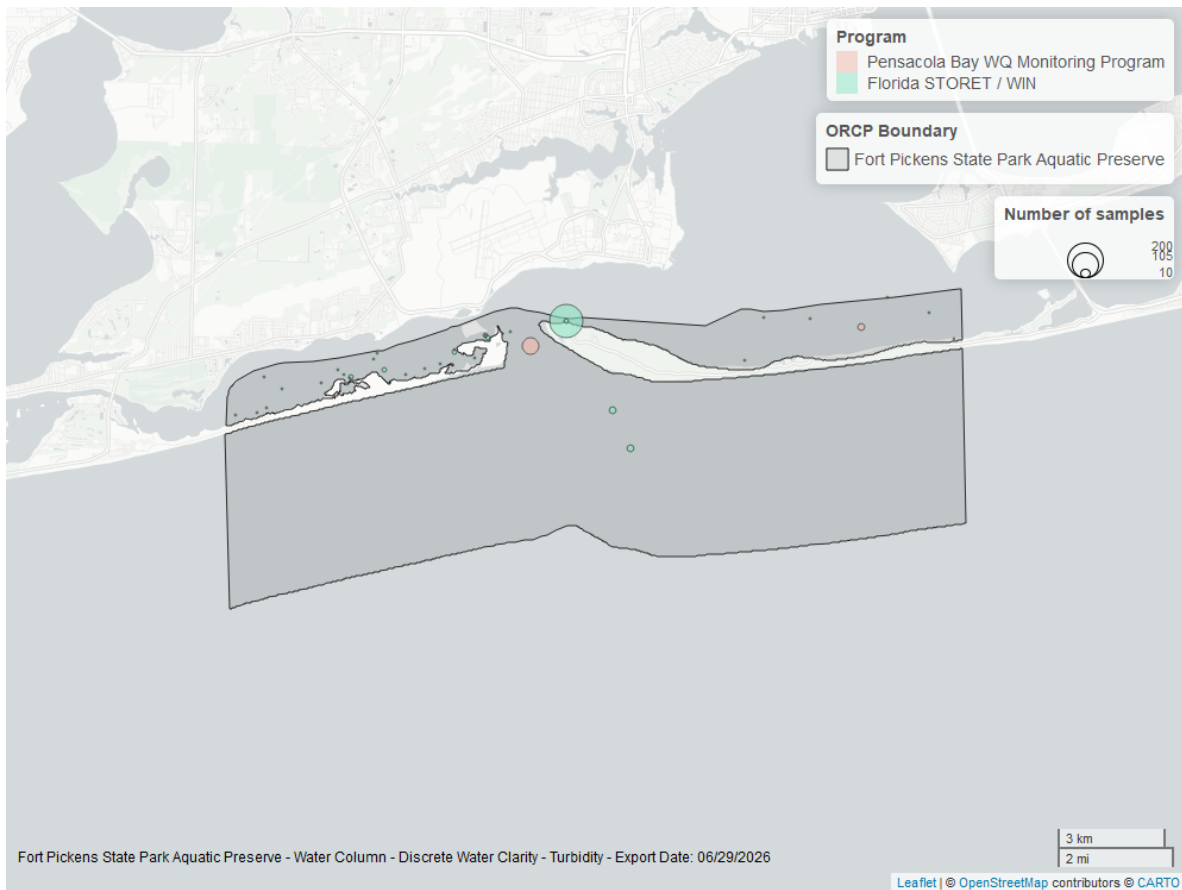


Figure 28: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

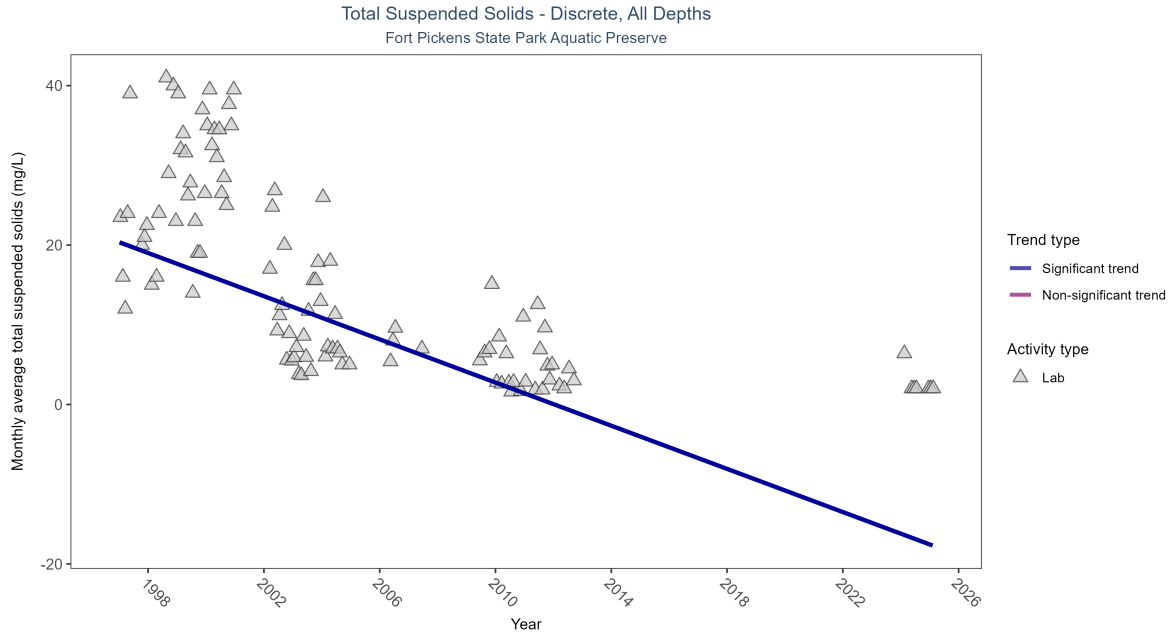


Figure 29: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 14: Monthly average total suspended solids decreased by 1.35 mg/L per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	264	15	1997 - 2025	9.2	-0.63671	20.36003	-1.35417	0

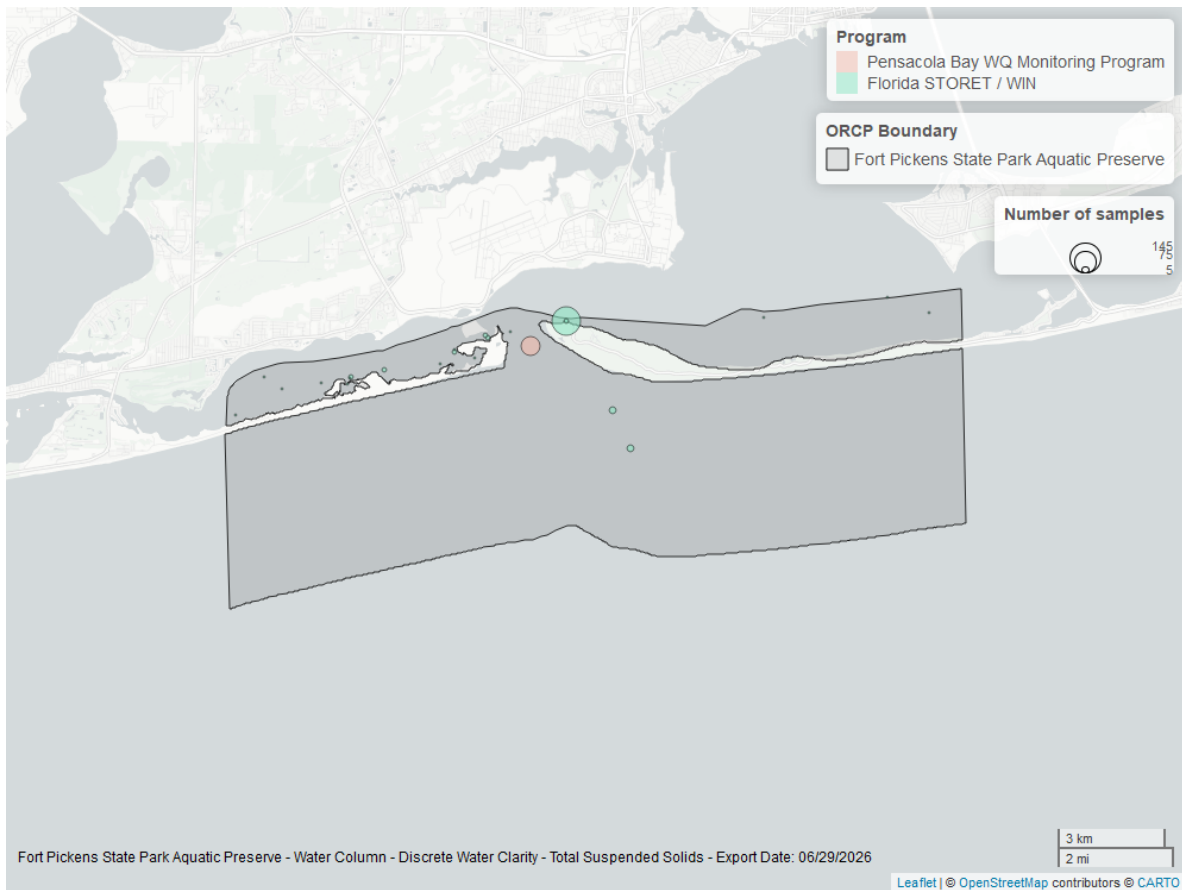


Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

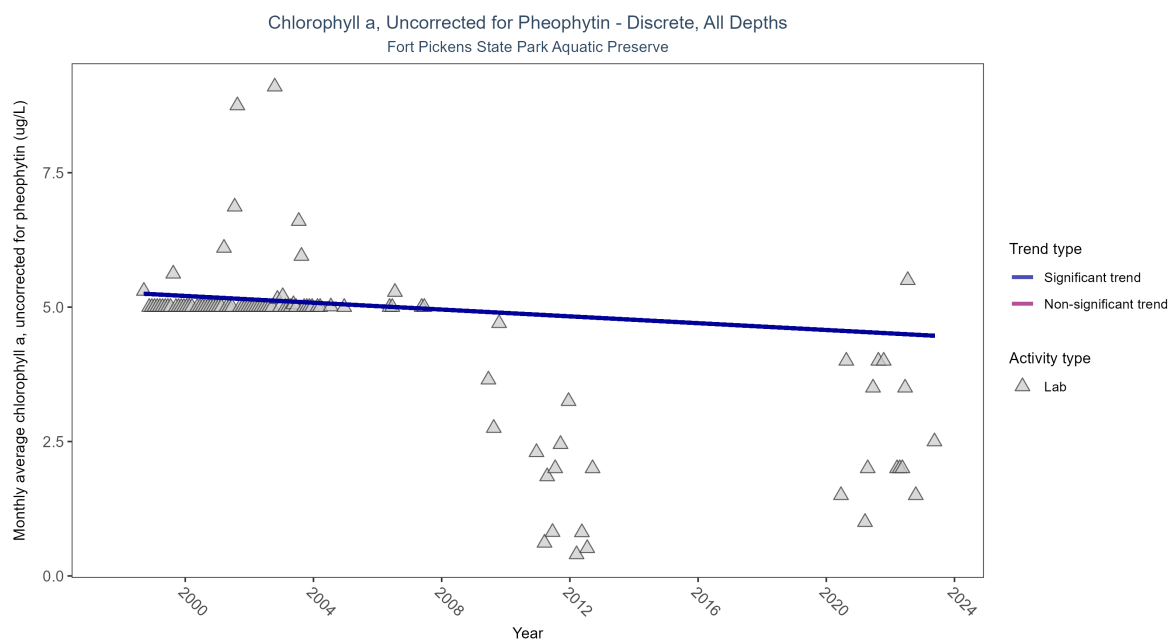


Figure 31: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 15: Monthly average chlorophyll a, uncorrected for pheophytin, decreased by 0.03 ug/L per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	224	17	1998 - 2023	5	-0.32141	5.27153	-0.03167	0

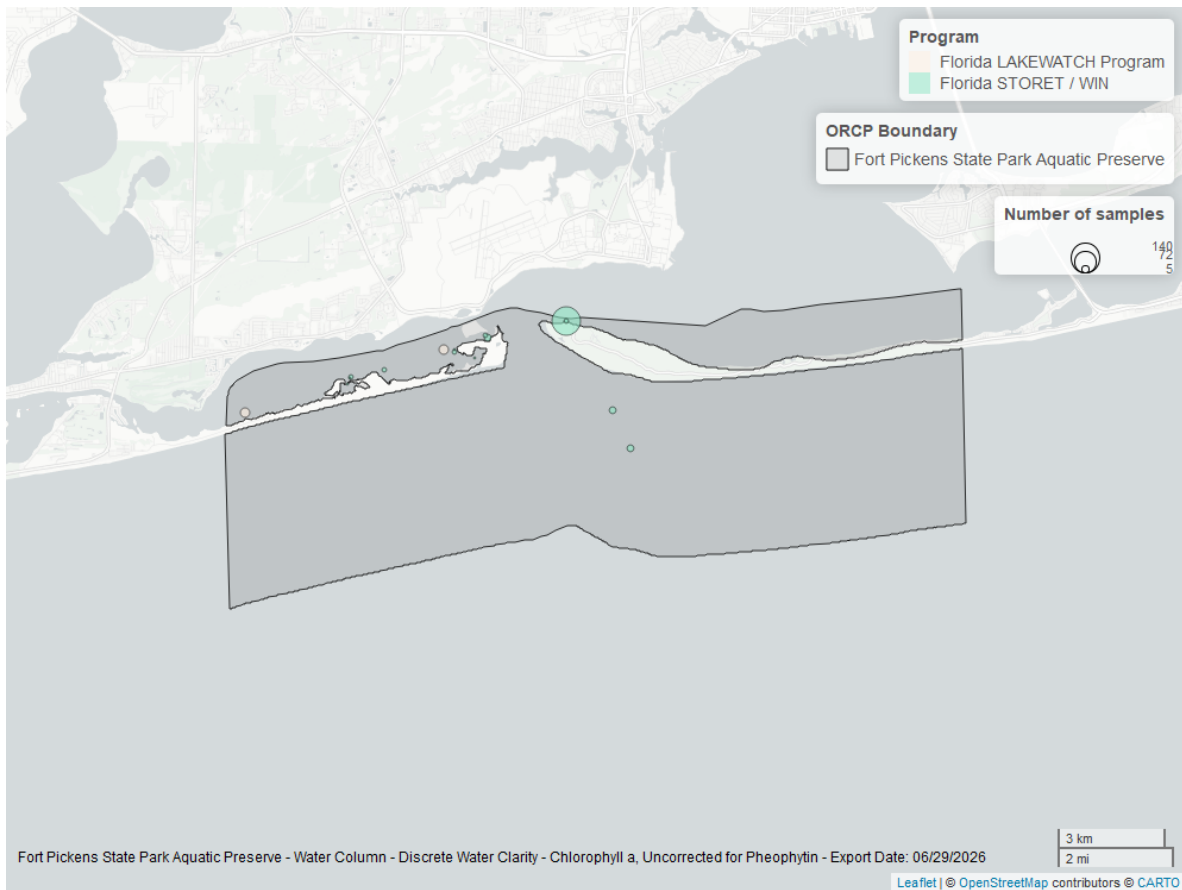


Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

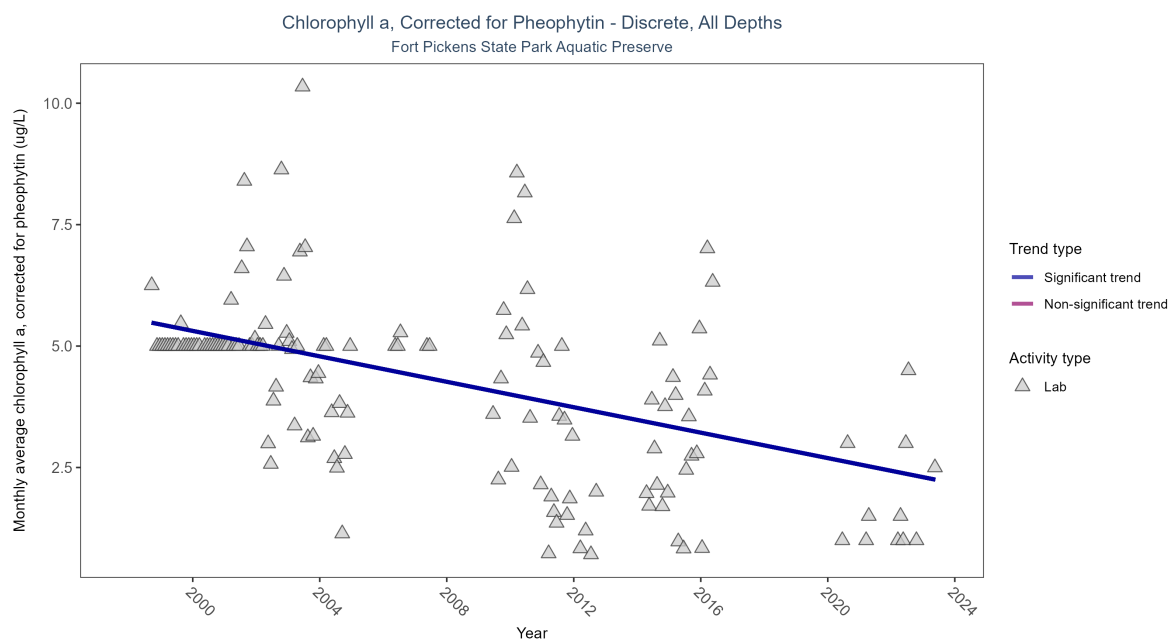


Figure 33: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 16: Monthly average chlorophyll a, corrected for pheophytin, decreased by 0.13 ugg/L per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	307	20	1998 - 2023	5	-0.3836	5.57382	-0.13091	0

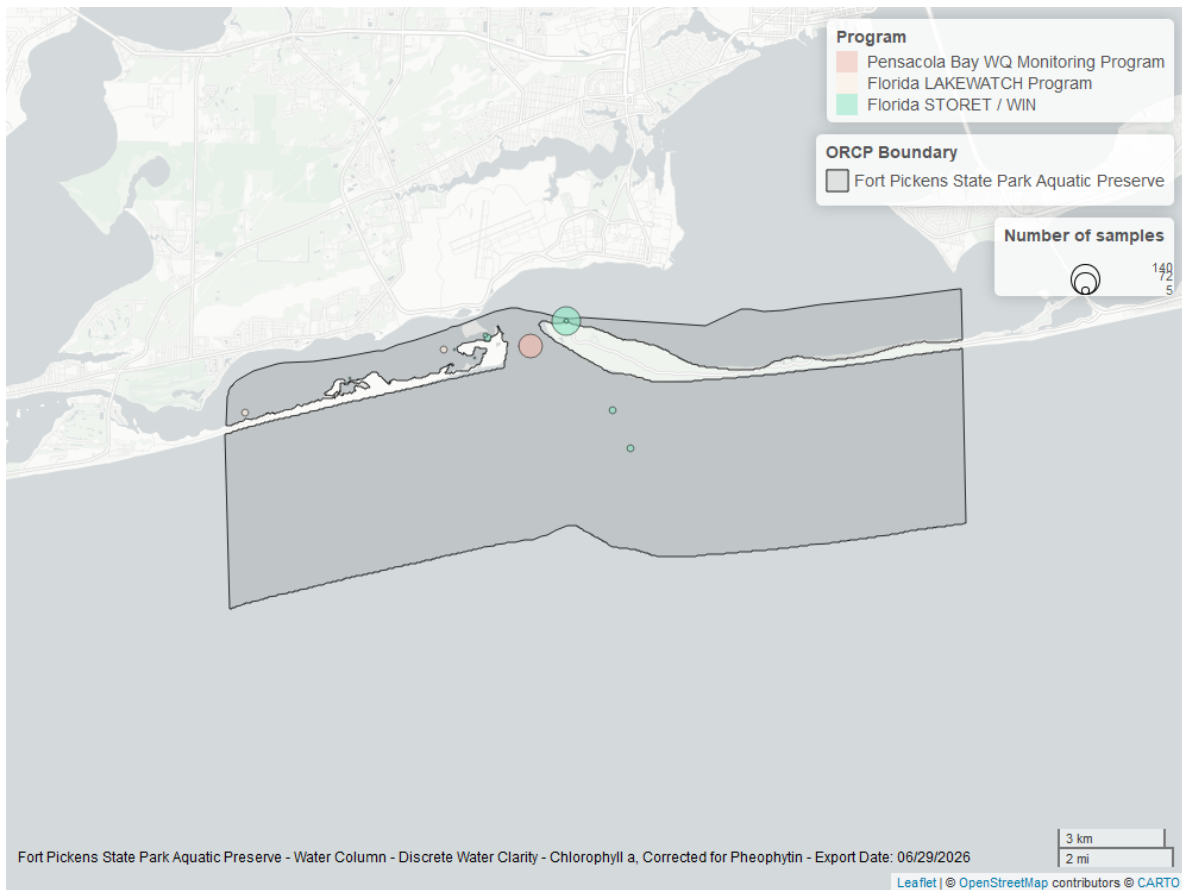


Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

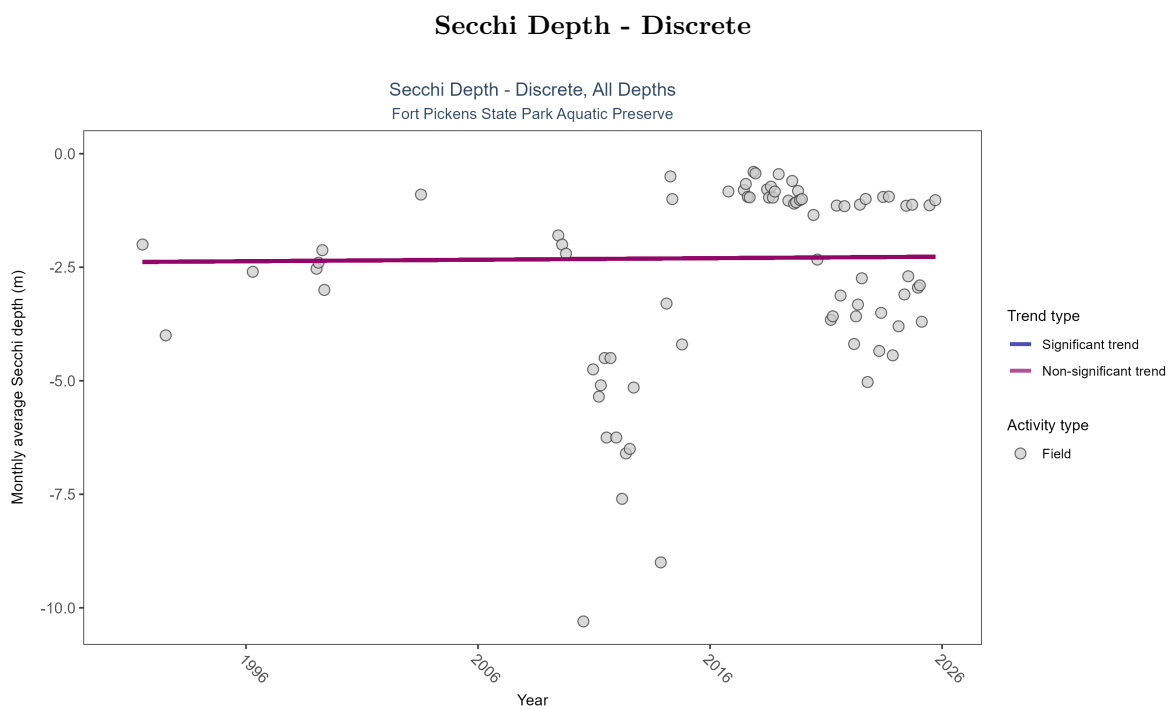


Figure 35: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 17: Secchi depth showed no detectable trend between 1991 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	591	21	1991 - 2025	-0.91441	-0.04892	-2.38525	0.00337	0.87

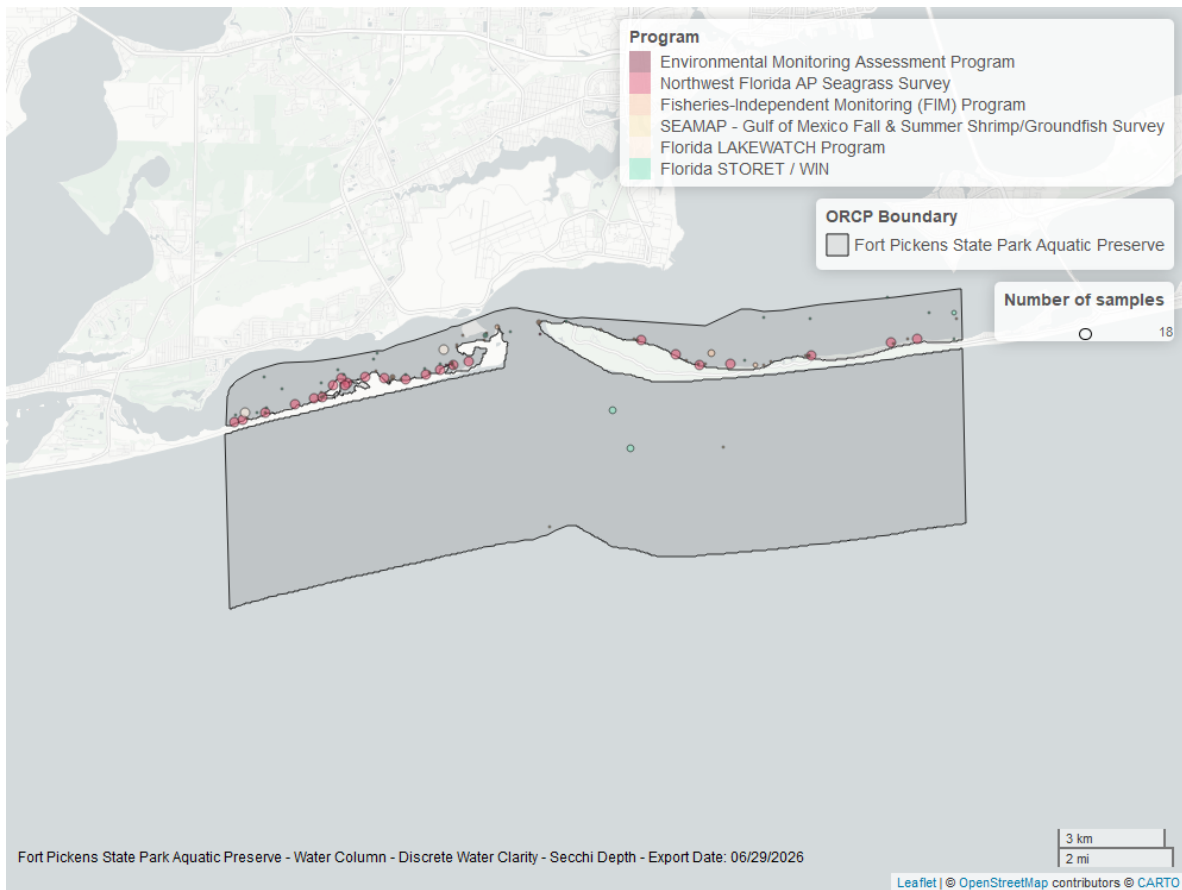


Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

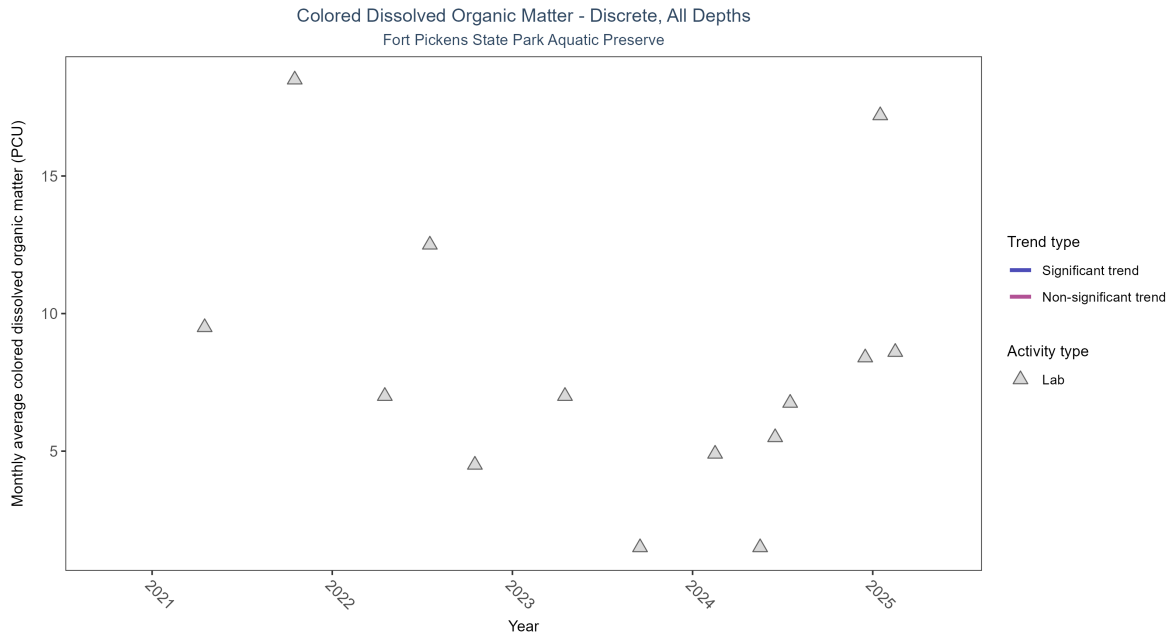


Figure 37: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 18: There was insufficient data to fit a model for colored dissolved organic matter.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data	23	5	2021 - 2025	7	-	-	-	-

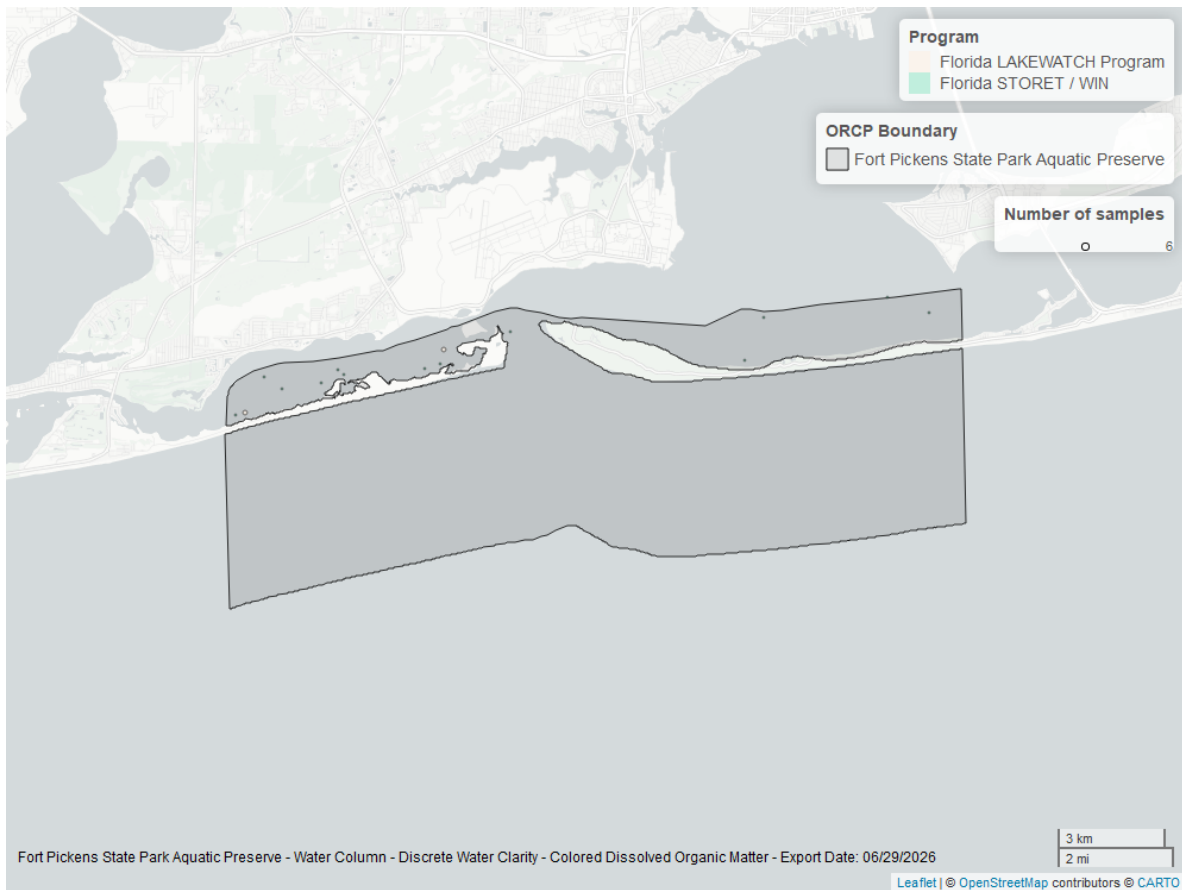


Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.